



Evaluation of the finite journal bearing characteristics, using the exact analytical solution of the Reynolds equation

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ABSTRACT

A very recent analysis of the exact analytical solution of the Reynolds equation for the lubrication of the plain journal bearing with finite length is used in the current work in order to evaluate the impedance forces of the fluid film and define them in closed form analytical expressions. The main journal bearing design parameters such as the eccentricity ratio, the location of the minimum fluid film thickness, the location of the maximum pressure and the stiffness and damping coefficients are evaluated under the recent analytical definition of the lubricant pressure distribution and presented in charts as functions of the Sommerfeld number, comparing them with the past and recent literature.

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1. Introduction

Journal bearings are used in all kind of rotating machinery and have the most important role in the dynamic behavior of the rotational shafts by being perhaps the only machine element able to suppress various exciting forces acting on the rotational elements. As rotating machines became large and multi-staged, while compactness, high speed, and high output have also been realized in recent years, not only has the bearing load increased, but also the magnitude and the variety of exciting forces. Therefore, the role and importance of journal bearings have increased tremendously. In particular, for the design of rotating machines with low vibration levels and high reliability, knowledge of the exact characteristic data of bearings, of the operational parameters, and especially of the stiffness and damping coefficients of oil films in bearings is essential.

The amount of reliable data now available to practical bearing design is explicitly based on numerical solutions and numerical procedures that nowadays, with the combination of high computational rates offered from the computers, can be judged sufficient for the reliability of bearing design. However, journal bearing design data under analytical methods are very few and limited in approximating solutions of the Reynolds equation for lubrication

of journal bearings. Since almost half a century, closed form expressions exist for the bearing operational parameters and characteristics but they are not reliable in the cases that the journal bearing is of finite length. The various numerical methods such as finite difference method, finite volume method, finite element method etc. gave the solution about the pressure distribution in a finite journal bearing and additionally, numerical methods using perturbations in the Reynolds equation or direct differentiation techniques yielded very useful data about the finite journal bearing characteristics. Under the consideration that the available data for the finite bearing design is obtained numerically, this work is considered to be a study of applying an analytical solution of the exact Reynolds equation for the plain full finite journal bearing in its analysis and design.

The analysis and design of the journal bearings is a concept considerably hindered in the numerical procedures for the solution of the Reynolds equation or in the necessity of employing side flow factors for treating bearings of finite length to diameter proportions. These necessities have been brought about because analytical they were available only for the so called infinitely long and infinitely short bearings. The side flow factors, even when they are properly employed, are based on partial arc bearings [1] and therefore cannot be strictly applicable to bearings of other arcs. Michell [2] and Muskat et al. [3] eliminated the need for side flow factors for the case of plain slider bearings by solving the fundamental equation for sliding pads of finite length to diameter ratio. The application of their solution to analysis and design of the pad type bearings has been presented by Raimondi and Boyd [4].

The case of the finite journal bearing cannot be dealt as the finite slider because of the more complicated film shape and the

Abbreviations: ODE, ordinary differential equation; PDE, Partial Differential Equation; BVP, Boundary Value Problem; PSM, Power Series Method; FDM, finite difference method; CFD, Computational Fluid Dynamics; FEM, Finite Elements Method; NRM, Newton–Raphson method; BC, boundary condition

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Nomenclature

P	resulting pressure of lubricant
u	pressure of the particular solution
g	pres. of the homogeneous solution
x	axial coordinate of the bearing
θ	angular coordinate of the bearing
θ^*	angular location of zero pressure
h	fluid film thickness
h^*	approximate fluid film thickness
R	journal radius

Ω	journal rotational speed
μ	lubricant dynamic viscosity
φ_0	attitude angle of the journal
c_r	bearing radial clearance
e	journal eccentricity
$\dot{e}$	journal eccentricity rate of change
ε	journal eccentricity ratio
$\dot{\varepsilon}$	journal eccentricity ratio rate of change
W	external bearing load
L_b	bearing/journal length
$c_1 - c_8$	constants of integration

addition of two more variables, the bearing arc and the position of the load line with respect to the bearing. The L/D ratio is one of the parameters upon which side flow factors depend and is combined also with other bearing parameters such as Sommerfeld number. However, the combinations for the side flow factors appear to have mathematical and experimental justification only in the case of the very short full journal bearing [5].

Cameron and Wood [6] published data for the full finite journal bearing for eccentricity ratios up to 0.8 which they obtained by relaxation methods. Sassenfeld and Walther [7] gave friction and film thickness information for both the 360° and 180° bearing. Data on finite bearings of special types has also been presented by Pinkus [8]. Raimondi and Boyd [9, Part I] solved the finite journal bearing fundamental equations numerically and they presented the results in very useful charts and tables that permit the determination of friction, film thickness, flow, temperature rise, load carrying capacity and other basic parameters for the design of a full finite journal bearing but also for bearings of arcs of 60° , 120° and 180° and for the widely used case of $L/D=1$. The corresponding results for journal bearings of various arcs and other L/D ratios were also presented from Raimondi and Boyd in a series of papers [10,11]. The numerical solution results supersede previous results [12] in which the same authors treated the finite journal bearing by the use of the side flow factors.

The research contributions in the analytical calculation of the pressure distribution in a journal bearing with finite length are not many and have not been made recently. Kingsbury [13] determined the pressure distribution by an experimental electrical analogy. Christopherson [14] determined the pressure distribution by utilizing the mathematical model of “relaxation”. Vogelpohl [15–17] achieved to give closed forms of the functions for the pressure distribution along both directions of the journal bearing, axial and circumferential. He also [15–17] assumed a partial solution of the Reynolds equation that corresponds to a “long bearing approximation” and then used the technique of separation of variables for the solution of the homogeneous Reynolds equation. By adding the two solutions, the pressure distribution was defined in closed form. Cameron and Wood [18] had extended the work of Christopherson [14] to show the effect of length to diameter ratio on eccentricity ratio, attitude angle and friction coefficient. In all cases, these solutions express natural phenomena in the oil film on the basis of Reynolds assumptions regarding lubrication, the most important assumption being that certain terms in the generalized Navier–Stokes equations for flow in a viscous fluid may be neglected.

The analytical treatment of the Reynolds equation for the finite bearing in order to obtain the lubricant pressure distribution function in a closed form analytical expression is the basis for the current work. The main assumption is that the lubricant is considered to be Newtonian, incompressible and to flow under laminar, isoviscous, and isothermal conditions. A path for an

analytical solution of the Reynolds equation was given in [15–17] by making an assumption of linear (instead of trigonometrical) fluid film thickness function. Other authors [19,20] used Bessel functions in order to solve the approximating Reynolds equation for the finite bearing with linear varying fluid film thickness. Recent works by the authors [21–23] use the exact fluid film function and thus an exact analytical solution for the pressure distribution in the finite journal bearing is obtained. The analytical solution is given in short in the current work, which is dedicated in the analytical expansion of the plain cylindrical finite journal bearing forces that are defined with direct integration of the pressure distribution function. The analytical expressions of the forces are used in order to obtain results about the main design parameters of a plain journal bearing such as loading capacity, minimum fluid film thickness location and the location (locus) of the maximum pressure.

The concept of the evaluation of the finite journal bearing characteristics of stiffness and damping is treated analytically since the analytical expansion of the finite journal bearing forces make this path feasible. In the available literature the numerical methods were used for the evaluation of the eight linear elastic and damping coefficients that determine the behaviour of the journal bearing working in steady state conditions. The classic fundamental works of Krämer [24], Glienicke [25], Lund [26], Lund and Thomsen [27], Someya [28], Sternlicht [29], Lang et al. [30], Gasch et al. [31] and others incorporate numerical methods for solving the Reynolds equation and extracting the stiffness and damping coefficients of the finite journal bearing while they present also experimental measurements on this complicated concept. The charts for the values of the stiffness and damping coefficients that are given in the literature mentioned above, together with many other basic contributions, are considered to be the reliable base for journal bearing system design and in some cases the results are presented in the charts of the current work also for the justification of the current evaluations. Closed form formulas and charts for the stiffness and damping coefficients of the short and long bearing are presented in [32–34].

By applying direct differentiation to the bearing forces functions, the stiffness and damping coefficients are obtained in closed form expressions and they are plotted in comparison to those obtained in classic literature. The Gumbel (half Sommerfeld) boundary condition for the pressure distribution is used to produce the results presented in the current work and unfortunately the comparison with the results given in literature is not always feasible since the Reynolds boundary condition has been used also widely. Although the numerical methods for solving the Reynolds equation and producing the bearing characteristics and design parameters are accurate, fast and efficient enough to design the journal bearings, the analytical methods for obtaining the corresponding results can be considered to be useful since the finite journal bearing impedance forces formulas can be used

directly in rotor dynamic calculation scripts, where the journal bearing solution is sometimes called numerous times during the calculations.

The current paper is divided in six sections. In Section 2 the short description of the exact analytical solution of the Reynolds equation for the finite journal bearing is presented. The path for obtaining the solution is extensively presented in [21] as well as in [22] but in this work only the final functions for the pressure definition are incorporated. However, the closed form expression for the pressure distribution under the Gumbel boundary condition is presented in the end of Section 2 and the pressure profile is plotted for some static cases of various eccentricity ratios in comparison with recent [36] and past [37] literature where numerical methods of CFD and FDM were applied. Additionally, the comparison incorporates results of the pressure profile under analytical solutions that assume linearly varying fluid film thickness function (approximate analytical) instead of trigonometric (exact analytical). The idea of linear fluid film thickness function was given by Vogelpohl [15], Michell [19] and Duffing [20] and considers sinusoidal or Bessel eigenfunctions for a Sturm–Liouville problem that appears during the path of the analytical solution. In the current work, the use of the approximate analytical solution using sinusoidal or Bessel eigenfunctions is made in order to give, in the following sections, the formulas of the fluid film forces much more compactly than those from Power Series Method (exact analytical). The application of the approximating fluid film thickness in the Reynolds equation and the derivation of the approximate analytical solution can be also found in [21].

In Section 3 the analytical derivation of the fluid film forces of the finite journal bearing is performed for the cases of the exact analytical solution using PSM and the approximate analytical solution using Bessel and sinusoidal eigenfunctions. After the definition of the forces formulas, the evaluation of some of the main design parameters of the journal bearing is performed in Section 4. The justification incorporates comparison with the numerical results from classic literature and with the approximate analytical results evaluated during this work. A finite difference method script was also used to solve the Reynolds equation and to offer numerical results for comparison that are not available in the literature under the current boundary condition (Gumbel). The numerical treatment using FDM and the numerical evaluation of the bearing characteristics under the FDM solution is presented in Appendix A. In Section 5, the stiffness and damping coefficients are defined with analytical closed form expressions under the exact analytical solution (using PSM) and the results for various bearing L/D ratios are compared to those from classic literature obtained numerically. The paper ends in Section 6 with the basic concluding remarks.

2. Short description of the exact analytical solution of the Reynolds equation

The problem of the lubrication of journal bearings with finite length was defined in [21] as the calculation of the pressure distribution of the Newtonian lubricant that is assumed to flow under laminar, isoviscous, and isothermal conditions in between the rotating journal and the static bearing, see Fig. 1. The journal of radius R and length L_b is assumed to be rotating with a constant rotational speed and to be constantly located in a point of eccentricity e with respect to the geometric centre of the bearing of radius $R+c_r$ and length L_b after an application of a virtual vertical load W as shown in Fig. 1. The “inputs” for the pressure evaluation are considered to be the eccentricity e and its rate of change $\dot{e}$. The journal and the bearing are supposed to be in parallel (aligned

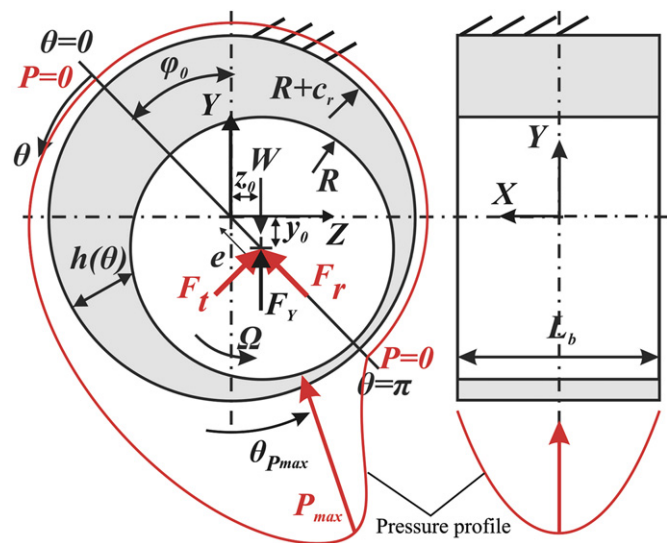


Fig. 1. The plain finite journal bearing scheme and the lubricant pressure distribution that is assumed to follow the Gumbel (half Sommerfeld) boundary condition in the results of the current study.

bearing) and the fluid film thickness h becomes a function of the unique independent parameter θ for a time moment of constant e and $\dot{e}$ which means that the function for the fluid film thickness is $h = c_r + e \cos(\theta)$ and its time derivative is $\partial h / \partial t = \dot{e} \cos(\theta)$. The dynamic viscosity of the lubricant is assumed to be constant and equal to μ through the entire control volume (notified with shadow in Fig. 1) that is defined from the bearing and the journal surfaces. The attitude angle of the journal is defined as ϕ_0 with respect to the vertical coordinate axis (see Fig. 1).

In this work, all the results are evaluated considering the Gumbel (half Sommerfeld) boundary condition for the fluid film pressure distribution. The definition of the pressure distribution function is presented shortly in this section, and in correspondence to [21], so as to present the role of the main functions that will be used in what follows. The detailed path for obtaining the solution is given in [21]. The closed form expression of the pressure distribution is used for the evaluation of the bearing impedance forces and the various operational and design parameters of the bearing.

The starting point is the equation of Reynolds which is expressed in Eq. (2.1) and its solution as in Eq. (2.2).

$$\frac{\partial}{\partial x} \left(\frac{h^3}{6\mu} \frac{\partial P(x, \theta)}{\partial x} \right) + \frac{1}{R^2} \frac{\partial}{\partial \theta} \left(\frac{h^3}{6\mu} \frac{\partial P(x, \theta)}{\partial \theta} \right) = \Omega \frac{\partial h}{\partial \theta} + 2 \frac{\partial h}{\partial t} \quad (2.1)$$

$$P(x, \theta) = \underbrace{u(x, \theta)}_{\text{particular}} + \underbrace{g(x, \theta)}_{\text{homogeneous}} \quad (2.2)$$

In order to evaluate the functions $u(x, \theta)$ and $g(x, \theta)$ a particular solution and the homogenous solution of Eq. (2.1) is needed. For doing so, we assume that $u(x, \theta)$ and $g(x, \theta)$ can be split in the following additive and multiplicative forms of Eqs. (2.3) and (2.4) respectively. The particular and the homogeneous solution of the Reynolds equation are of the form of Eqs. (2.3) and (2.4) and this assumption leads to four Sturm–Liouville problems for the correspondent functions $\varphi(x, \theta)$, $\psi(x)$, $f(\theta)$ and $m(x)$.

$$u(x, \theta) = \varphi(\theta) + \psi(x) \quad (2.3)$$

$$g(x, \theta) = f(\theta)m(x) \quad (2.4)$$

We are looking for a solution where the independent variables can be split in the above form of Eq. (2.3). If we substitute Eq. (2.3)

in Eq. (2.1) after some calculations Eq. (2.5) is obtained.

$$-\frac{d^2\psi(x)}{dx^2} = \frac{1}{R^2} \frac{d^2\varphi(\theta)}{d\theta^2} - \frac{3e\sin(\theta)}{R^2(c_r + e\cos(\theta))} \frac{d\varphi(\theta)}{d\theta} + \frac{6\mu e\Omega\sin(\theta)}{(c_r + e\cos(\theta))^3} - \frac{12\mu\dot{e}\cos(\theta)}{(c_r + e\cos(\theta))^3} \quad (2.5)$$

$$c_3 = \frac{6R^2\mu\left(\pi\left(2(c_r - e)^2\sqrt{-c_r^2 + e^2}\dot{e} + 3c_re^3\Omega\left(\text{Log}\left(-\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right) - \text{Log}\left(\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right)\right)\right) + 3c_re^3\Omega\arccos\left(\frac{-c_r}{e}\right)\left(\text{Log}\left(-\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right) - \text{Log}\left(\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right)\right)\right)}{e(2c_r^2 + e^2)\left(\pi\left(\text{Log}\left(-\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right) - \text{Log}\left(\frac{c_r + e}{\sqrt{-c_r^2 + e^2}}\right)\right) + \arccos\left(\frac{-c_r}{e}\right)\left(-\text{Log}\left(-\frac{\sqrt{-c_r^2 + e^2}}{\sqrt{e}\sqrt{c_r + e}}\right) + \text{Log}\left(\frac{\sqrt{-c_r^2 + e^2}}{\sqrt{e}\sqrt{c_r + e}}\right)\right)\right)} \quad (2.14)$$

By inspecting the above equation one observes that the right hand side is a function of θ while the left hand side is a function of x only. So, the equality will be feasible only when both sides are equal to the same constant C . From the latter equation we obtain two equations for the functions $\varphi(\theta)$ and $\psi(x)$. These are ordinary differential equations written as in Eqs. (2.6) and (2.7).

$$-\frac{d^2\psi(x)}{dx^2} = C \quad (2.6)$$

$$\frac{1}{R^2} \frac{d^2\varphi(\theta)}{d\theta^2} - \frac{3e\sin(\theta)}{R^2(c_r + e\cos(\theta))} \frac{d\varphi(\theta)}{d\theta} + \frac{6\mu e\Omega\sin(\theta)}{(c_r + e\cos(\theta))^3} - \frac{12\mu\dot{e}\cos(\theta)}{(c_r + e\cos(\theta))^3} = C \quad (2.7)$$

Eq. (2.6) can be solved directly to give Eq. (2.8). The constants c_1, c_2 are the arbitrary constants of integration calculated from the boundary conditions. For solving Eq. (2.7) one observes that we talk about a linear ordinary differential equation with non-constant coefficients where the unknown function $\varphi(\theta)$ is not present explicitly. So, by setting $z(\theta)$ as in Eq. (2.9), instead of Eq. (2.7) we may solve the following linear ordinary differential equation defined in Eq. (2.10) of the first order for the function $z(\theta)$ [40].

$$\psi(x) = -Cx^2/2 + c_1x + c_2 \quad (2.8)$$

$$\frac{d\varphi(\theta)}{d\theta} = z(\theta) \quad (2.9)$$

$$\frac{1}{R^2} \frac{dz(\theta)}{d\theta} - \frac{3e\sin(\theta)}{R^2(c_r + e\cos(\theta))} z(\theta) + \frac{6\mu e\Omega\sin(\theta)}{(c_r + e\cos(\theta))^3} - \frac{12\mu\dot{e}\cos(\theta)}{(c_r + e\cos(\theta))^3} = C \quad (2.10)$$

$$z(\theta) = \left(e^{\int \left(\frac{3e\sin(\theta)}{c_r + e\cos(\theta)} d\theta \right)} \left(c_3 + \int \left(e^{-\int \left(\frac{3e\sin(\theta)}{c_r + e\cos(\theta)} d\theta \right)} \left(-\frac{6\mu R^2 e\sin(\theta)}{(c_r + e\cos(\theta))^3} + \frac{12\mu R^2 \dot{e}\cos(\theta)}{(c_r + e\cos(\theta))^3} + C \right) d\theta \right) \right) \quad (2.11)$$

All the integrals in Eq. (2.11) can be evaluated using direct closed form expressions. By evaluating the integrals that appear in the expression for $z(\theta)$ one can integrate the outcome one time and obtain the function $\varphi(\theta)$, Eq. (2.12). The sum of $\varphi(\theta)$ and $\psi(x)$ renders classes of particular solutions by choosing values for the constants C, c_1, c_2, c_3, c_4 . By choosing $C=0$ we obtain Eqs. (2.12) and (2.13).

$$\psi(x) = c_1x + c_2 \quad (2.12)$$

$$\varphi(\theta) = c_4 + \frac{1}{2} \left(-\frac{2B(-18c_re^2R^2\mu\Omega + (2c_r^2 + e^2)c_3)}{(-c_r^2 + e^2)^{5/2}} \right) + \frac{1}{2} \left(\frac{3e(2c_r^2R^2\mu\Omega - c_rc_3)\sin(\theta)}{(c_r^2 - e^2)^2(c_r + e\cos(\theta))} \right)$$

$$+ \frac{12(-c_r^2 + e^2)\dot{e}R^2\mu - e^2(6c_rR^2\mu\Omega - c_3)\sin(\theta)}{(-c_r^2 + e^2)(c_r + e\cos(\theta))^2} \quad (2.13)$$

After applying the Gumbel (half Sommerfeld) boundary condition for the pressure distribution, the function $\varphi(\theta)$ is defined with the constants of integration to be given in Eqs. (2.14) and (2.15).

$$c_4 = -6\dot{e}R^2\mu/e(c_r + e)^2 \quad (2.15)$$

The solution for $\psi(x)$ is chosen to be the trivial one for reasons that are explained in detail in [21] but in the current work the path for obtaining the expressions of the impedance forces includes also the general assumption of non-trivial solution for $\psi(x)$ [21]. In general the solution function of $\psi(x)$ is defined as in Eq. (2.16) but since the constant C in the Sturm–Liouville problem of $\psi(x)$ is set $C=0$ then the constants c_1 and c_2 become also zero, Eq. (2.17).

$$\psi(x) = -Cx^2/2 + c_1x + c_2 \quad (2.16)$$

$$\psi(x) = 0 \quad (2.17)$$

The evaluation of $g(x, \theta)$ of Eq. (2.2) is performed to give the generic set of solutions for the homogeneous Reynolds equation, namely for Eq. (2.1) with the right hand terms equal to zero, as defined in Eq. (2.18).

$$\frac{(c_r + e\cos(\theta))^3}{6\mu} \frac{\partial^2 g(x, \theta)}{\partial x^2} - \frac{3(c_r + e\cos(\theta))^2 e\sin(\theta)}{6\mu R^2} \frac{\partial g(x, \theta)}{\partial \theta} + \frac{(c_r + e\cos(\theta))^3}{6\mu R^2} \frac{\partial^2 g(x, \theta)}{\partial \theta^2} = 0 \quad (2.18)$$

By using this assumption in Eq. (2.4) one will arrive at the following outcome of Eq. (2.19).

$$-\frac{m''(x)}{m(x)} = \frac{1}{R^2} \frac{f''(\theta)}{f(\theta)} - \frac{1}{R^2} \frac{3e\sin(\theta)}{(c_r + e\cos(\theta))} \frac{f'(\theta)}{f(\theta)} \quad (2.19)$$

By inspecting the latter equation one realizes that the left hand is a function of x while the right a function of θ only. So, in order this equality to be feasible both sides should be equal to the same constant, say λ . We thus obtain Eqs. (2.20) and (2.21) [21].

$$-\frac{m''(x)}{m(x)} = \lambda \quad (2.20)$$

$$\frac{1}{R^2} \frac{f''(\theta)}{f(\theta)} - \frac{1}{R^2} \frac{3e\sin(\theta)}{(c_r + e\cos(\theta))} \frac{f'(\theta)}{f(\theta)} = \lambda \quad (2.21)$$

The Sturm–Liouville problem for the function $f(\theta)$ is here written as in Eq. (2.22) with $p(\theta)=h(\theta)^3$, $q(\theta)=0$ and $r(\theta)=R^2h(\theta)^3$. There is no direct solution and it is confronted using the method of Power Series [38,39]. The eigenfunctions $f_i(\theta)$ are defined as in Eq. (2.23) with the Power Series constants to be defined in Eq. (2.24) as function of the eigenvalues k_i presented in Eq. (2.25).

$$(p(\theta)f'(\theta))' - q(\theta)f(\theta) - \lambda r(\theta)f(\theta) = 0, \quad f(\theta) = 0 \text{ and } f'(\theta^*) = 0 \quad \text{with} \quad \lambda = -k^2 \quad (2.22)$$

$$f_i(\theta) = \sum_{n=0}^{\infty} \delta_n \cos(\theta)^n, \quad i = 1, 2, \dots, \quad (2.23)$$

$$\delta_{-1} = 0(\alpha n^2 + \alpha n - (2 + 3\gamma_i)\alpha)\delta_{n-1} + (\beta n^2)\delta_n$$

$$-(xn^2 + 4xn + 3\alpha)\delta_{n+1} - (\beta n^2 + 3n\beta + 2\beta)\delta_{n+2} = 0, \quad n = 0, 1, 2, \dots \quad (2.24)$$

$$\alpha = e/3, \quad \beta = c_r/3, \quad \gamma_i = 2k_i R^2 \quad (2.25)$$

The forth Sturm–Liouville problem, for the function $m(x)$, yields the eigenfunctions $m_i(x)$ defined in Eq. (2.26) [21].

$$m_i(x) = \frac{e^{L_b k_i/2}}{1 + e^{L_b k_i}} (e^{k_i x} + e^{-k_i x}), \quad i = 1, 2, \dots \quad (2.26)$$

The resulting eigenfunctions $f_i(\theta)$ are normalized in Eq. (2.27), using the constants defined in Eq. (2.28).

$$\bar{f}_i(\theta) = \eta_i f_i(\theta), \quad i = 1, 2, \dots \quad (2.27)$$

$$\eta_i = 1 / \int_0^{\theta^*} r(\theta) f_i(\theta)^2 d\theta, \quad i = 1, 2, \dots \quad (2.28)$$

The homogenous solution is defined as in Eq. (2.29) with the constants σ_i to be given from Eq. (2.30).

$$g(x, \theta) = \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(x)) \quad (2.29)$$

$$\sigma_i = \int_0^{\theta^*} (\varphi(\theta) r(\theta) \bar{f}_i(\theta)) d\theta, \quad i = 1, 2, \dots \quad (2.30)$$

The resulting pressure is given in the closed form expression of Eq. (2.31). The pressure distribution in a bearing of $L_b/D=1$ is evaluated and presented in Fig. 2 for variable cases of eccentricity ratio and $\varepsilon=0$ in comparison with numerical results from the literature [21]. The first ten eigenfunctions $\bar{f}_i(\theta)$ and $m_i(x)$ were used in the calculation of pressure.

$$P(x, \theta) = \varphi(\theta) + \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(x)), \quad \theta \in [0, \pi], \quad x \in [-L_b/2, L_b/2] \quad (2.31)$$

3. Analytical expressions for the finite bearing fluid film forces

Using the closed form expression of Eq. (2.31) for the pressure distribution in a plain finite bearing, the fluid film resultant impedance forces are calculated in this section. The journal center is assumed to be located in the point defined in Fig. 1 with a displacement of y_0 in the vertical direction and of z_0 in the horizontal direction. The eccentricity and the attitude angle are expressed in Eq. (3.1).

$$e = \sqrt{y_0^2 + z_0^2}, \quad \phi_0 = \arctan(z_0/y_0) \quad (3.1)$$

The journal is not always supposed to be under static equilibrium condition. For this reason the eccentricity and the attitude angle rate

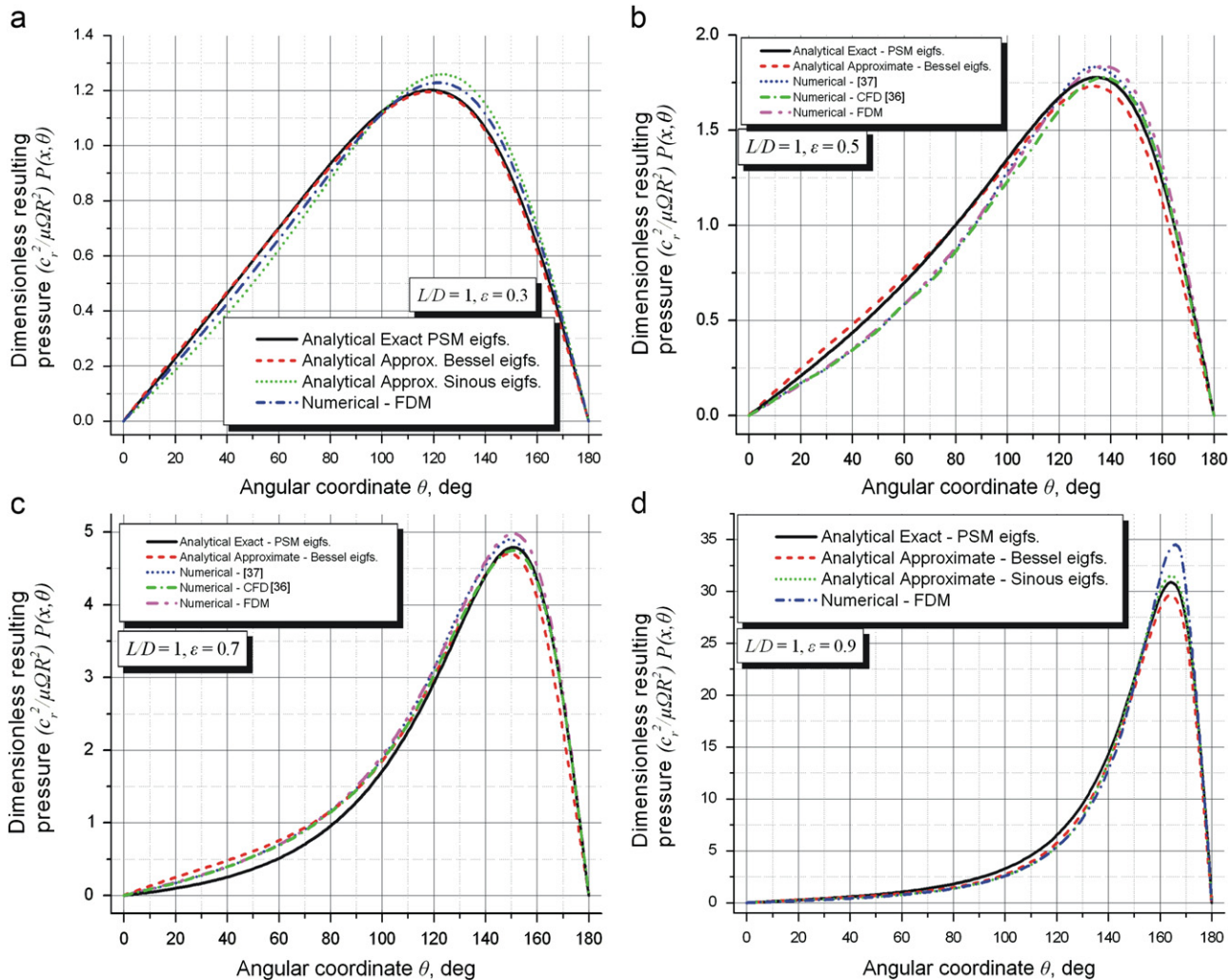


Fig. 2. The resulting pressure profile through the circumferential direction of a bearing with $L_b/D=1$ using the exact analytical solution, approximate analytical solutions, and numerical methods, for the relative eccentricity ratio of (a) $\varepsilon=0.3$, (b) $\varepsilon=0.5$, (c) $\varepsilon=0.7$ and (d) $\varepsilon=0.9$.

of change are incorporated in the analysis. The eccentricity rate of change is already presented in the initial definition of the Reynolds equation Eq. (2.1). The eccentricity rate of change and the attitude angle rate of change are defined in Eq. (3.2).

$$\dot{e} = (z_0\dot{z}_0 + y_0\dot{y}_0)/e, \quad \dot{\phi}_0 = (y_0\dot{z}_0 - z_0\dot{y}_0)/e^2 \quad (3.2)$$

The fluid film forces in the radial and the tangential direction F_r and F_t (see Fig. 1) are obtained using the formulas of Eqs. (3.3) and (3.4) correspondingly. Using the expression for the fluid film pressure distribution (Eq. (2.31)) the radial and tangential components can be expressed as the sum of the forces yielded from the particular solution (Eq. (2.3)) $F_{r,p}$ and $F_{t,p}$, and the forces yielded from the homogeneous solution (Eq. (2.4)) $F_{r,h}$ and $F_{t,h}$ as described in Eqs. (3.3) and (3.4). Towards generality, the fluid film pressure function is integrated in angular domain with an upper limit of θ^* . The angle θ^* describes the angular location of zero pressure, that in Gümpel (half Sommerfeld) boundary condition is set $\theta^* = \pi$. A detailed study for the values of θ^* is presented in [9] but in this work it is preferred to follow the simplest assumption in order the resulting force formulas to be expressed in the most compact edition.

$$\begin{aligned} F_r &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} P(x, \theta) R \cos(\theta) dx d\theta = \underbrace{\int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} u(x, \theta) R \cos(\theta) dx d\theta}_{= F_{r,p}} \\ &\quad \text{(due to particular solution)} \\ &\quad + \underbrace{\int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} g(x, \theta) R \cos(\theta) dx d\theta}_{= F_{r,h}} \\ &\quad \text{(due to homogenous solution)} \end{aligned} \quad (3.3)$$

$$\begin{aligned} F_t &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} P(x, \theta) R \sin(\theta) dx d\theta = \underbrace{\int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} u(x, \theta) R \sin(\theta) dx d\theta}_{= F_{t,p}} \\ &\quad \text{(due to particular solution)} \\ &\quad + \underbrace{\int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} g(x, \theta) R \sin(\theta) dx d\theta}_{= F_{t,h}} \\ &\quad \text{(due to homogenous solution)} \end{aligned} \quad (3.4)$$

In the following subsections the resultant forces F_r and F_t are expressed in closed form expressions by defining separately the components of the particular and homogenous solution.

3.1. Evaluation of the forces $F_{r,p}$ and $F_{t,p}$ due to the particular solution

The part of the pressure distribution defined by the particular solution $u(x, \theta) = \varphi(\theta) + \psi(x)$ is integrated through the journals space domain as in Eqs. (3.5) and (3.6) in order to give the forces $F_{r,p}$ and $F_{t,p}$. All the bounded integrals presented during the evaluation happen to have direct closed form expressions and the result is obtained using the expressions.

$$\begin{aligned} F_{r,p} &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} u(x, \theta) R \cos(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} (\varphi(\theta) + \psi(x)) R \cos(\theta) dx d\theta \end{aligned}$$

$$\begin{aligned} &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \varphi(\theta) R \cos(\theta) dx d\theta \\ &\quad + \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \psi(x) R \cos(\theta) dx d\theta \end{aligned} \quad (3.5)$$

$$\begin{aligned} F_{t,p} &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} u(x, \theta) R \sin(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} (\varphi(\theta) + \psi(x)) R \sin(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \varphi(\theta) R \sin(\theta) dx d\theta \\ &\quad + \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \psi(x) R \sin(\theta) dx d\theta \end{aligned} \quad (3.6)$$

Since the solution for $\varphi(\theta)$ is derived through the integration of $z(\theta)$, defined also in Eqs. (3.7) and (3.8), the forces $F_{r,p}$ and $F_{t,p}$ can be expressed as in Eqs. (3.9) and (3.10)

$$\varphi(\theta) = \int z(\theta) d\theta + c_4 \quad (3.7)$$

$$\begin{aligned} z(\theta) &= e^{\int \left(\frac{3e\sin(\theta)}{c_r + e\cos(\theta)} \right) d\theta} \left(c_3 + \int \left(e^{-\int \left(\frac{3e\sin(\theta)}{c_r + e\cos(\theta)} \right) d\theta} \left(-\frac{6\Omega\mu R^2 e\sin(\theta)}{(c_r + e\cos(\theta))^3} \right. \right. \right. \\ &\quad \left. \left. \left. + \frac{12\Omega\mu R^2 \dot{e}\cos(\theta)}{(c_r + e\cos(\theta))^3} + \underbrace{C}_{=0} \right) d\theta \right) \right) \\ &= \frac{c_3 + 6eR^2\mu\Omega\cos(\theta) + 12\dot{e}R^2\mu\Omega\sin(\theta)}{(c_r + e\cos(\theta))^3} \end{aligned} \quad (3.8)$$

$$\begin{aligned} F_{r,p} &= R \int_0^{\theta^*} \varphi(\theta) (\sin(\theta))' d\theta \int_{-L_b/2}^{L_b/2} 1 dx \\ &\quad + R \int_0^{\theta^*} \cos(\theta) d\theta \int_{-L_b/2}^{L_b/2} \psi(x) dx = RL_b \varphi(\theta) \sin(\theta) \\ &\quad - RL_b \underbrace{\int_0^{\theta^*} z(\theta) \sin(\theta) d\theta}_{I_1} + R \left[\underbrace{-Cx^2/2 + c_1x + c_2}_{=0} \right]_{-L_b/2}^{L_b/2} [\cos(\theta)]_0^{\theta^*} \end{aligned} \quad (3.9)$$

$$\begin{aligned} F_{t,p} &= R \int_0^{\theta^*} \varphi(\theta) (-\cos(\theta))' d\theta \int_{-L_b/2}^{L_b/2} 1 dx \\ &\quad + R \int_0^{\theta^*} \sin(\theta) d\theta \int_{-L_b/2}^{L_b/2} \psi(x) dx = -RL_b \varphi(\theta) \cos(\theta) \\ &\quad + RL_b \underbrace{\int_0^{\theta^*} z(\theta) \cos(\theta) d\theta}_{I_2} + R \left[\underbrace{-Cx^2/2 + c_1x + c_2}_{=0} \right]_{-L_b/2}^{L_b/2} [\sin(\theta)]_0^{\theta^*} \end{aligned} \quad (3.10)$$

The integrals I_1 and I_2 appeared in Eqs. (3.9) and (3.10) are evaluated as in Eqs. (3.11) and (3.12) and with the use also of Eq. (2.13) for $\varphi(\theta)$, the forces $F_{r,p}$ and $F_{t,p}$ obtain their final expressions.

$$\begin{aligned} I_1 &= \frac{12\dot{e}R^2\mu\Omega \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2})}{(-c_r^2 + e^2)^{3/2}} \\ &\quad + \frac{6R^2\mu\Omega(-c_r^2 + e^2 + c_r\dot{e}\sin(\theta))}{e(-c_r^2 + e^2)(c_r + e\cos(\theta))} \\ &\quad + \frac{c_3 - 6R^2c_r\mu\Omega + 12\dot{e}R^2\mu\Omega\sin(\theta)}{2e(c_r + e\cos(\theta))^2} \end{aligned} \quad (3.11)$$

$$I_2 = \frac{-3 \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2})e(-c_3 c_r + 2(c_r^2 + 2e^2)R^2 \mu \Omega)}{(-c_r^2 + e^2)^{5/2}} + \frac{24(c_r^2 + e^2)^2 \dot{e} R^2 \mu \Omega (c_r + e \cos(\theta)) + 2c_r e^2 (2c_3 c_r^2 + c_3 e^2 - 18c_r e^2 R^2 \mu \Omega) \sin(\theta) + e^3 (c_3 (c_r^2 + 2e^2) + 6c_r (c_r^2 - 4e^2) R^2 \mu \Omega) \sin(2\theta)}{4e^2 (c_r^2 + e^2)^2 (c_r + e \cos(\theta))^2} \quad (3.12)$$

3.2. Evaluation of the forces $F_{r,h}$ and $F_{t,h}$ due to the homogeneous solution

The part of the pressure distribution defined by the homogeneous solution $g(x, \theta)$ has now to be integrated through the journals space domain. As discussed in Section 2, the function $f(\theta)$ has no direct solution and thus the Power Series Method is used in order to define it. The integration of the series function of Eq. (2.23) will lead to the expression of the resulting forces using an infinite sum. Additionally to this way of expressing $f(\theta)$, Vogelpohl [15] and Duffing [20] present the way of deriving an approximate analytical solution for $f(\theta)$ by approximating the trigonometric fluid film thickness function $h = c_r + e \cos(\theta)$ with a linear fluid film thickness function $h^* = (c_r + e) - 2e\theta/\pi$ (see also [21,22]). The approximate analytical solution for $f(\theta)$ can be then expressed much more simply using sinusoidal expressions for the eigenfunctions $f_i(\theta)$ such as $f_i(\theta) = \sin(i\theta)$, $i = 1, 2, \dots$. The fluid film forces corresponding to the homogeneous solution $g(x, \theta) = f(\theta)m(x)$ are calculated in Section 3.2.1 as the exact fluid film forces and in Section 3.2.2 as the approximating fluid film forces.

3.2.1. Evaluation of the forces $F_{r,h}$ and $F_{t,h}$ using the method of Power Series (exact forces)

The eigenfunctions of the Sturm–Liouville problem for $f(\theta)$ are expressed in Eq. (3.13) with the use also of Eqs. (3.14) and (3.15).

$$f_i(\theta) = \sum_{n=0}^{\infty} \delta_n \cos(\theta)^n, \quad i = 1, 2, \dots, \quad (3.13)$$

$$\delta_{-1} = 0(\alpha n^2 + \alpha n - (2 + 3\gamma_i)\alpha)\delta_{n-1} + (\beta n^2)\delta_n - (\alpha n^2 + 4\alpha n + 3\alpha)\delta_{n+1} - (\beta n^2 + 3n\beta + 2\beta)\delta_{n+2} = 0 \quad (3.14)$$

$$\alpha = e/3, \quad \beta = c_r/3, \quad \gamma_i = 2k_i R^2 \quad (3.15)$$

The normalized eigenfunctions $\bar{f}_i(\theta)$ are derived using Eqs. (2.27) and (2.28) and the forces $F_{r,h}$ and $F_{t,h}$ are initially defined in closed form expressions as in Eqs. (3.16) and (3.17).

$$\begin{aligned} F_{r,h} &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} g(x, \theta) R \cos(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} \sigma_i \bar{f}_i(\theta) m_i(x) R \cos(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \sum_{n=0}^{\infty} \delta_n \cos(\theta)^n m_i(x) \right) R \cos(\theta) dx d\theta \end{aligned} \quad (3.16)$$

$$\begin{aligned} F_{t,h} &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} g(x, \theta) R \sin(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} \sigma_i \bar{f}_i(\theta) m_i(x) R \sin(\theta) dx d\theta \\ &= \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \sum_{n=0}^{\infty} \delta_n \cos(\theta)^n m_i(x) \right) R \sin(\theta) dx d\theta \end{aligned} \quad (3.17)$$

After the evaluation of the integrals appeared in the equations above, the fluid film forces $F_{r,h}$ and $F_{t,h}$ can be expressed as in

Eqs. (3.18) and (3.19).

$$\begin{aligned} F_{r,h} &= R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \int_0^{\theta^*} \cos(\theta)^{n+1} d\theta \right) \int_{-L_b/2}^{L_b/2} m_i(x) dx \right) \\ &= R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \int_0^{\theta^*} \cos(\theta)^{n+1} d\theta \right) \times \left[\frac{e^{(L_b-2x/2)k_i} (-1 + e^{2k_i x})}{(1 + e^{L_b k_i}) k_i} \right]_{-L_b/2}^{L_b/2} \right) \end{aligned} \quad (3.18)$$

$$\begin{aligned} F_{t,h} &= R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \int_0^{\theta^*} \cos(\theta)^n \sin(\theta) d\theta \right) \int_{-L_b/2}^{L_b/2} m_i(x) dx \right) \\ &= R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \int_0^{\theta^*} \cos(\theta)^n \sin(\theta) d\theta \right) \times \left[\frac{e^{(L_b-2x/2)k_i} (-1 + e^{2k_i x})}{(1 + e^{L_b k_i}) k_i} \right]_{-L_b/2}^{L_b/2} \right) \end{aligned} \quad (3.19)$$

By evaluating the integrals that appear in the equations above, the forces $F_{r,h}$ and $F_{t,h}$ are expressed as in Eqs. (3.20) and (3.21) correspondingly.

$$\begin{aligned} F_{r,h} &= 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \left(\frac{\pi^{3/2} \sec(n\pi)}{2 \Gamma(-n/2)} \left(-\frac{2^{2+n} \sqrt{\pi}}{\Gamma(2+n)} \right. \right. \right. \right. \\ &\quad \left. \left. \left. + \frac{(-1 + (-1)^n \cos(n\pi)) \csc(n\pi/2) \Gamma(-n/2)}{\Gamma((3+n)/2)} \right) \right) \right) \frac{\tanh(L_b k_i/2)}{k_i} \right) \end{aligned} \quad (3.20)$$

$$F_{t,h} = 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \frac{1 + (\cos(\theta^*))^{n+1}}{(1+n)\theta^*} \right) \frac{\tanh(L_b k_i/2)}{k_i} \right) \quad (3.21)$$

By setting $\theta^* = \pi$ (half Sommerfeld boundary condition) the forces $F_{r,h}$ and $F_{t,h}$ are expressed as in Eqs. (3.22) and (3.23) correspondingly.

$$\begin{aligned} F_{r,h} &= 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \left(\frac{\pi^{3/2} \sec(n\pi)}{2 \Gamma(-n/2)^2} \left(-\frac{2^{2+n} \sqrt{\pi}}{\Gamma(2+n)} \right. \right. \right. \right. \\ &\quad \left. \left. \left. + \frac{(-1 + (-1)^n \cos(n\pi)) \csc(n\pi/2) \Gamma(-n/2)}{\Gamma((3+n)/2)} \right) \right) \right) \times \frac{\tanh(L_b k_i/2)}{k_i} \right) \end{aligned} \quad (3.22)$$

$$F_{t,h} = 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \frac{1 + (-1)^n}{(1+n)} \right) \frac{\tanh(L_b k_i/2)}{k_i} \right) \quad (3.23)$$

The resulting forces are evaluated and plotted in Fig. 3 as a function of the value of terms of the infinite series in order to show that they present a quick convergence. Even though it is not feasible to prove the convergence of the series (which is a physically reasonable result since they express a force) there is strong evidence from the numerical calculations that the series converge very fast. The solution does not go to infinity and there is also a good behaviour in the sense that small differences lead to small changes in the outcome.

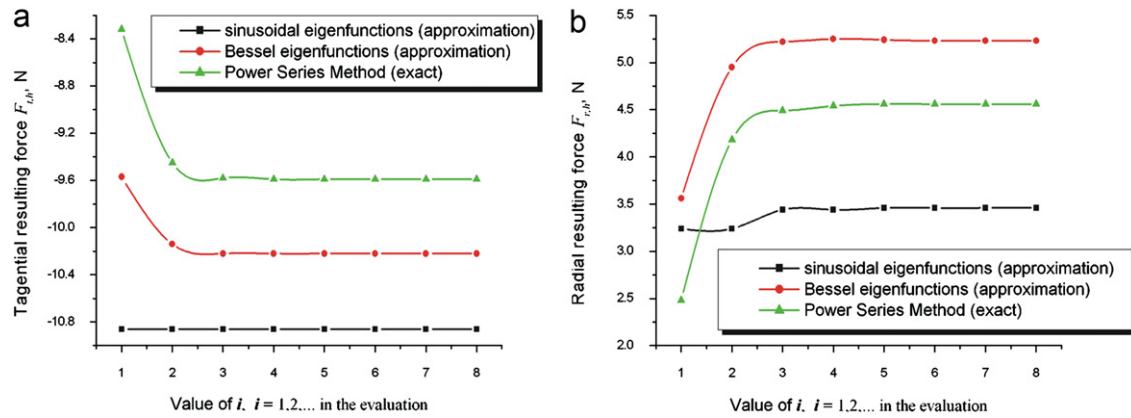


Fig. 3. Example of the values (and the convergence) of the analytical resulting forces given by the homogenous solution as a function of the value of $i=1,2,\dots$ (number of sum), for the three different definitions of the eigenfunctions (approximating analytical using 1. sinusoidal and 2. Bessel eigfs. and 3. exact analytical using eigfs. obtained with PSM). $L_b/D=1$, $\varepsilon=0.7$ and $\theta^*=\pi$.

3.2.2. Evaluation of $F_{r,h}$ and $F_{t,h}$ using the approximating analytical solution (approximate forces)

3.2.2.1. Using sinusoidal eigenfunctions for the Sturm–Liouville problem of $f(\theta)$. Using the eigenfunctions $f_i(\theta)=\sin(i\theta)$, $i=1,2,\dots$, and after their normalization in Eq. (3.24), the forces $F_{r,h}$ and $F_{t,h}$ are defined in closed form expressions in Eqs. (3.25) and (3.26), while the Eq. (3.27) is used for the definition of the constants σ_i .

$$\bar{f}_i(\theta) = \frac{1}{\sqrt{\pi}} \sin(i\theta), \quad i=1,2,\dots, \quad (3.24)$$

$$F_{r,h} = \frac{1}{\sqrt{\pi}} \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} (\sigma_i \sin(i\theta) m_i(x)) R \cos(\theta) dx d\theta \\ = \frac{R}{\sqrt{\pi}} \sum_{i=1}^{\infty} \left(\sigma_i \int_0^{\theta^*} \sin(i\theta) \cos(\theta) d\theta \int_{-L_b/2}^{L_b/2} m_i(x) dx \right) \quad (3.25)$$

$$F_{t,h} = \frac{1}{\sqrt{\pi}} \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} (\sigma_i \sin(i\theta) m_i(x)) R \sin(\theta) dx d\theta \\ = \frac{R}{\sqrt{\pi}} \sum_{i=1}^{\infty} \left(\sigma_i \int_0^{\theta^*} \sin(i\theta) \sin(\theta) d\theta \int_{-L_b/2}^{L_b/2} m_i(x) dx \right) \quad (3.26)$$

$$\sigma_i = \frac{1}{\sqrt{\pi}} \int_0^{\pi} -\varphi(\theta) \sin(i\theta) m_i \left(\frac{L_b}{2} \right) d\theta, \quad i=1,2,\dots \quad (3.27)$$

The bounded integrals presented in Eqs. (3.25) and (3.26) happen to have direct expressions of definition and the resulting forces can be expressed simply as in Eqs. (3.28) and (3.29).

$$F_{r,h} = \frac{R}{\sqrt{\pi}} \sum_{i=1}^{\infty} \left(\sigma_i \frac{i - i \cos(\theta^*) \cos(i\theta^*) + \sin(\theta^*) \sin(i\theta^*) \tanh(L_b k_i/2)}{-1 + i^2} \frac{1}{k_i} \right) \quad (3.28)$$

$$F_{t,h} = \frac{R}{\sqrt{\pi}} \sum_{i=1}^{\infty} \left(\sigma_i \frac{-i \cos(i\theta^*) \sin(\theta^*) + \cos(\theta^*) \sin(i\theta^*) \tanh(L_b k_i/2)}{-1 + i^2} \frac{1}{k_i} \right) \quad (3.29)$$

For $i=1$ the nominators and the denominators at both expressions in Eqs. (3.28) and (3.29) become zero; for this reason their limits are used in the numerical evaluation and the forces obtain

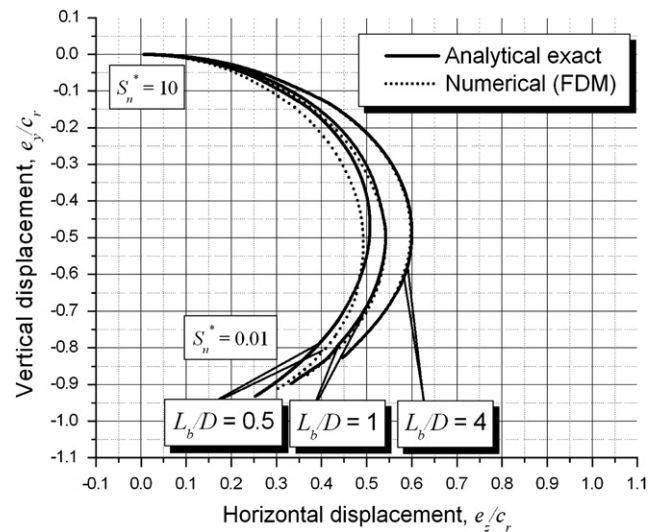


Fig. 4. Journal equilibrium trajectory through Sommerfeld number values, evaluated using the exact analytical solution (using PSM) for different L_b/D ratios.

the formulas of Eqs. (3.30) and (3.31).

$$F_{r,h} = \frac{R \sigma_1 \sin(\theta^*)^2 \tanh(L_b k_1/2)}{2\sqrt{\pi} k_1} \\ + \frac{R}{\sqrt{\pi}} \sum_{i=2}^{\infty} \left(\sigma_i \frac{i - i \cos(\theta^*) \cos(i\theta^*) + \sin(\theta^*) \sin(i\theta^*) \tanh(L_b k_i/2)}{-1 + i^2} \frac{1}{k_i} \right) \quad (3.30)$$

$$F_{t,h} = \frac{R \sigma_1 (\theta^* - \cos(\theta^*) \sin(\theta^*)) \tanh(L_b k_1/2)}{2\sqrt{\pi} k_1} \\ + \frac{R}{\sqrt{\pi}} \sum_{i=2}^{\infty} \left(\sigma_i \frac{-i \cos(i\theta^*) \sin(\theta^*) + \cos(\theta^*) \sin(i\theta^*) \tanh(L_b k_i/2)}{-1 + i^2} \frac{1}{k_i} \right) \quad (3.31)$$

The forces obtained by the last two formulas in Eqs. (3.30) and (3.31) are evaluated and plotted in Fig. 4 as a function of i in order to obtain a remark for the convergence of the infinite sum series. As shown in Fig. 4 the infinite sum series of Eqs. (3.30) and (3.31)

converge very fast and only the first ten or less eigenfunctions can be incorporated in the evaluation of the forces. Additionally, the infinite sum presented in Eq. (3.31) can be totally neglected when $\theta^* = \pi$ since it is equal to zero (see Fig. 3). By setting $\theta^* = \pi$ the final formulas for the approximate $F_{r,h}$ and $F_{t,h}$ are defined as in Eqs. (3.32) and (3.33).

$$F_{r,h} = \frac{R}{\sqrt{\pi}} \sum_{i=2}^{\infty} \left(\frac{\sigma_i(i + i\cos(i\pi)) \tanh(L_b k_i/2)}{(-1 + i^2) k_i} \right) \quad (3.32)$$

$$F_{t,h} = \frac{R\pi\sigma_1 \tanh(L_b k_1/2)}{2\sqrt{\pi} k_1} \quad (3.33)$$

The resulting forces are evaluated and plotted in Fig. 3 as a function of the value of terms of the infinite series in order to show the convergence that they present.

3.2.2.2. Using Bessel eigenfunctions for the Sturm–Liouville problem of $f(\theta)$ (linear film thickness $h^*(\theta)$). When the fluid film thickness is assumed to be a linear function of θ [15,19,20], the Sturm–Liouville problem defined by Eq. (2.22) has direct solution and its eigenfunctions are given by Eq. (3.34).

$$f_i(\theta) = -\frac{\text{BesselJ}(1, ((c_r + e)k_i\pi R/2e) - k_i R\theta)}{c_r\pi + e\pi - 2e\theta} \times \frac{\text{BesselY}(1, -(c_r + e)k_i\pi R/2e)}{\text{BesselJ}(1, (c_r + e)k_i\pi R/2e)} + \frac{\text{BesselY}(1, -(c_r + e)k_i\pi R/2e + k_i R\theta)}{c_r\pi + e\pi - 2e\theta}, \quad i = 1, 2, \dots \quad (3.34)$$

The eigenvalues k_i are calculated as the roots of the characteristic equation of the Sturm–Liouville problem (2.22) defined in Eq. (3.35).

$$-\text{BesselJ}\left(1, k\pi R - \frac{(c_r + e)k\pi R}{2e}\right) \text{BesselY}\left(1, \frac{-(c_r + e)k\pi R}{2e}\right) - \text{BesselJ}\left(1, \frac{(c_r + e)k\pi R}{2e}\right) \text{BesselY}\left(1, k\pi R - \frac{(c_r + e)k\pi R}{2e}\right) = 0 \quad (3.35)$$

The eigenfunctions defined above become normalized using Eqs. (3.36) and (3.37). The weight function $r(\theta)$ (see also

of the constants σ_i .

$$\bar{f}_i(\theta) = \eta_i f_i(\theta), \quad i = 1, 2, \dots, \quad (3.36)$$

$$\eta_i = 1 / \int_0^{\theta^* = \pi} r(\theta) f_i(\theta)^2 d\theta, \quad i = 1, 2, \dots, \quad (3.37)$$

$$F_{r,h} = \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(x)) R \cos(\theta) dx d\theta \quad (3.38)$$

$$F_{t,h} = \int_0^{\theta^*} \int_{-L_b/2}^{L_b/2} \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(x)) R \sin(\theta) dx d\theta \quad (3.39)$$

$$\sigma_i = \int_0^{\theta^*} -\varphi(\theta) R^2 ((c_r + e) - 2e\theta/\pi)^3 \bar{f}_i(\theta) d\theta, \quad i = 1, 2, \dots \quad (3.40)$$

The resulting forces are evaluated and plotted in Fig. 3 as a function of the value of terms of the infinite series in order to show the convergence that they present. Additionally, in order to show that the infinite sum series converge, a study is presented in Appendix A.

3.3. Expressions for the exact fluid film forces F_Y and F_Z of the finite bearing

Using the fluid film forces derived from the particular and the homogeneous solution of Reynolds equation, the resulting radial and tangential components of the fluid film forces are expressed as in Eqs. (3.41) and (3.42). In what follows, the results that correspond to the exact analytical solution of the Reynolds equation are derived using the following force formulas in Eqs. (3.41) and (3.42). The Gumpel boundary condition for the pressure profile was used by setting $\theta^* = \pi$.

$$F_r = RL_b \left(c_4 + \frac{1}{2} \left(-\frac{2 \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2})(-18c_r e^2 R^2 \mu \Omega + (2c_r^2 + e^2)c_3)}{(-c_r^2 + e^2)^{5/2}} + \frac{3e(2c_r^2 R^2 \mu \Omega - c_r c_3) \sin(\theta)}{(c_r^2 + e^2)^2 (c_r + e \cos(\theta))} \right) + \frac{12(-c_r^2 + e^2) \dot{e} R^2 \mu - e^2 (6c_r R^2 \mu \Omega - c_3) \sin(\theta)}{(-c_r^2 e + e^3)(c_r + e \cos(\theta))^2} \right) \sin(\theta) - RL_b \left(\frac{12 \dot{e} R^2 \mu \Omega \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2})}{(-c_r^2 + e^2)^{3/2}} + \frac{6R^2 \mu \Omega (-c_r^2 + e^2 + c_r \dot{e} \sin(\theta))}{e(-c_r^2 + e^2) \left(c_r + e \cos(\theta) \right) + \frac{c_3 - 6R^2 c_r \mu \Omega + 12 \dot{e} R^2 \mu \Omega \sin(\theta)}{2e(c_r + e \cos(\theta))^2}} \right) + 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \left(\frac{\pi^{3/2} \sec(n\pi)}{2 \Gamma(-n/2)^2} \left(-\frac{2^{2+n} \sqrt{\pi}}{\Gamma(2+n)} \right) \right) \right) \frac{\tanh(L_b k_i/2)}{k_i} \right) \right) \quad (3.41)$$

$$F_t = -RL_b \left(c_4 + \frac{1}{2} \left(-\frac{2 \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2})(-18c_r e^2 R^2 \mu \Omega + (2c_r^2 + e^2)c_3)}{(-c_r^2 + e^2)^{5/2}} + \frac{3e(2c_r^2 R^2 \mu \Omega - c_r c_3) \sin(\theta)}{(c_r^2 - e^2)^2 (c_r + e \cos(\theta))} + \frac{12(-c_r^2 + e^2) \dot{e} R^2 \mu - e^2 (6c_r R^2 \mu \Omega - c_3) \sin(\theta)}{(-c_r^2 e + e^3)(c_r + e \cos(\theta))^2} \right) \right) \cos(\theta) + RL_b \frac{-3 \operatorname{arctanh}((c_r - e)\tan(\theta/2)/\sqrt{-c_r^2 + e^2}) e (-c_3 c_r + 2(c_r^2 + 2e^2) R^2 \mu \Omega)}{(-c_r^2 + e^2)^{5/2}} + RL_b \frac{24(c_r^2 - e^2)^2 \dot{e} R^2 \mu \Omega (c_r + e \cos(\theta)) + 2c_r e^2 (2c_3 c_r^2 + c_3 e^2 - 18c_r e^2 R^2 \mu \Omega) \sin(\theta) + e^3 (c_3 (c_r^2 + 2e^2) + 6c_r (c_r^2 - 4e^2) R^2 \mu \Omega) \sin(2\theta)}{4e^2 (c_r^2 - e^2)^2 (c_r + e \cos(\theta))^2} + 2R \sum_{i=1}^{\infty} \left(\sigma_i \eta_i \left(\sum_{n=0}^{\infty} \delta_n \frac{1 + (-1)^n}{(1+n)} \right) \frac{\tanh(L_b k_i/2)}{k_i} \right) \right) \quad (3.42)$$

Eq. (2.22) now becomes $r(\theta) = -R^2(h^*)^3 = -R^2((c_r + e) - 2e\theta/\pi)^3$. The forces $F_{r,h}$ and $F_{t,h}$ are then defined in closed form expressions in Eqs. (3.38) and (3.39), while Eq. (3.40) is used for the definition

The radial and tangential components are used in Eq. (3.43) to give the resulting bearing forces F_Y and F_Z with respect to the stationary Cartesian coordinate system (see Fig. 1) by setting

$$\cos(\phi_0) = y_0/e \text{ and } \sin(\phi_0) = z_0/e.$$

$$F_Y = F_r \frac{y_0}{e} + F_t \frac{z_0}{e}, F_Z = F_t \frac{y_0}{e} - F_r \frac{z_0}{e} \quad (3.43)$$

4. Analytical evaluation of the finite journal bearing operational characteristics

The main operational and design parameters of the plain finite journal bearing are evaluated in this section under the use of the exact analytical solution and the two approximate analytical solutions of the Reynolds equation presented in the sections before. The Gümbel (half Sommerfeld) boundary condition is used in the pressure profile evaluation and the analytical expressions of the forces that were obtained in Sections 3.2.1, 3.2.2.1 and 3.2.2.2 and are used here to produce the results.

The journal equilibrium locus for various values of the Sommerfeld number is evaluated and plotted in Fig. 4 for various cases of bearing L/D ratio, using the exact analytical solution. The corresponding numerical result using FDM (see Appendix A) is presented also in Fig. 4. The Sommerfeld number $S_n^* = 1/2\pi (R/c_r)^2 (\mu L_b(2R)\Omega/W)$ varies from 0.01 (heavy loaded bearing) to 10 (light loaded bearing) and as it can be seen in Fig. 4 the agreement is very

good especially in the cases of L/D ratio of 1 and 4. A more detailed analysis in the locus of the equilibrium position is made by evaluating independently the eccentricity ratio and the attitude angle of the journal with the analytical and the numerical solutions.

The eccentricity ratio is evaluated with the exact analytical solution and is plotted as a function of the Sommerfeld number S_n^* in Fig. 5 in comparison to numerical results from the literature using FDM, FEM and CFD. In Fig. 5 it is seen that the agreement in eccentricity ratio is on the desirable domain, presenting a maximum discrepancy of only 0.05 in the some specific values of Sommerfeld number and only with one of the three numerical methods that are presented. Comparing with the numerical result using CFD, the eccentricity ratio under the exact analytical solution is almost in perfect agreement in all L/D ratios.

The results of the approximate solutions regarding eccentricity ratio are evaluated and plotted in comparison to the exact analytical result in Fig. 6. It can be seen that the agreement is very good presenting a maximum discrepancy of less than 0.05 in the case of L/D ratio of 0.25. The convergence between the analytical solutions is even better in L/D ratios of 1 and 4. The approximate analytical solution using Bessel eigenfunctions is presented to be closer to this of using sinusoidal eigenfunctions.

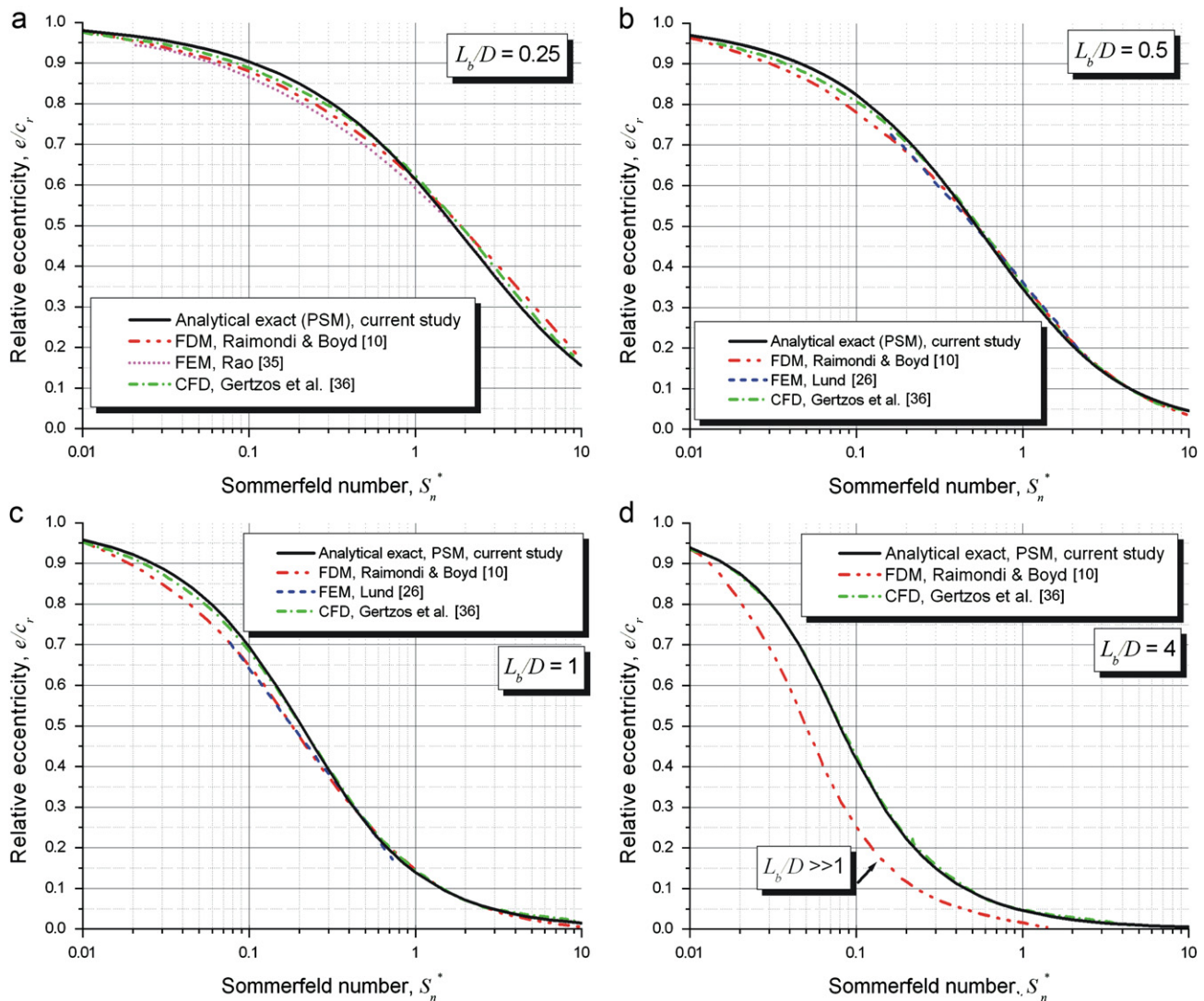


Fig. 5. Evaluation of the eccentricity ratio as a function of Sommerfeld number for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$ using the exact analytical solution and comparison with results from numerical solutions.

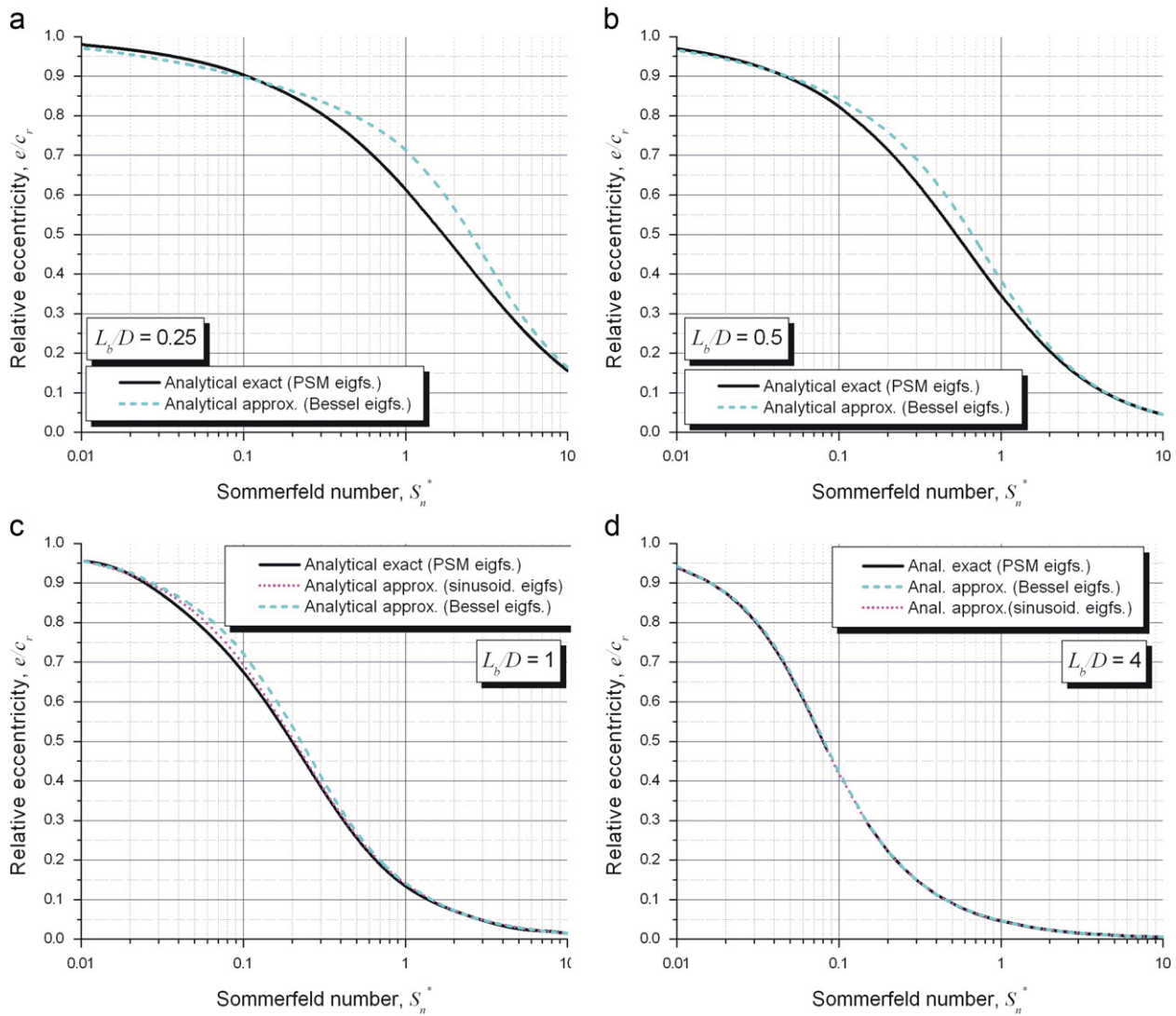


Fig. 6. Evaluation of the eccentricity ratio as a function of Sommerfeld number for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$ using the exact analytical solution and comparison with the approximate analytical solutions using sinusoidal and Bessel eigenfunctions.

Another important parameter of operation and design, this of attitude angle, or location of minimum fluid film thickness, is evaluated using the analytical solutions (exact and approximate) and plotted in Fig. 7 in comparison to numerical solutions from the literature. As notified in Fig. 7, considering the different solutions, the attitude angle under different evaluations is presented also with convergence, and the eccentricity ratio does also, as shown in Figs. 5 and 6. Thus, it can be said that these two basic parameters of journal bearing operation can be reliably evaluated even with the approximate analytical solutions using the more compact versions of fluid film forces instead of those of exact analytical forces that are presented to be more complicated.

In what follows the angular location of the maximum lubricant pressure is evaluated with the analytical solutions and compared with this of FDM presented in Appendix A. The Gümbel boundary condition for the pressure profile is used and the location of the maximum pressure is evaluated in the bearings axial centre where $x=0$. The evaluation is performed by solving Eq. (4.1) using a Newton–Raphson method. The first 10 eigenfunctions are used in the infinite sum presented in the function of the pressure profile. The sum $\theta_{pmax} + \varphi_0$ (see Fig. 1) is the outcome plotted in Fig. 8, incorporating both the attitude angle and the angle of the

maximum pressure in the π film.

$$\left. \frac{\partial \theta}{\partial \theta} \left(\varphi(\theta) + \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(\theta)) \right) \right|_{\theta = \theta_{pmax}} = 0 \quad (4.1)$$

The angle of the maximum pressure exists in the domain of the 3rd quadrant (see Fig. 1) since $\theta_{pmax} + \varphi_0$ is yielded to be from 180° to 205° . The L/D ratio appears to have an influence in the location of the maximum pressure. In a lightly loaded short bearing (e.g., $S_n^* = 10$) of L/D ratio of 0.25 the angle of maximum pressure appears to be at 203° while for a long bearing of L/D ratio of 4 the value $\theta_{pmax} + \varphi_0$ tends almost at 180° . In the case of a heavy loaded bearing (e.g., $S_n^* = 0.01$) the angle of maximum pressure changes from 185° in the short bearing, to almost 195° in the long bearing. The discrepancy between the results obtained by the exact analytical solution and those by FDM numerical solution (see Appendix A) and CFD solution [36] has a maximum of almost 5° in the case of the short bearing (L/D ratio of 0.25) while in the case of long bearing (L/D ratio of 4) almost no discrepancies are notified. The intermediate cases of L/D ratios of 1 and 0.5 present discrepancy of almost 2.5° and 4° correspondingly between the exact analytical and the FDM or CFD solution.

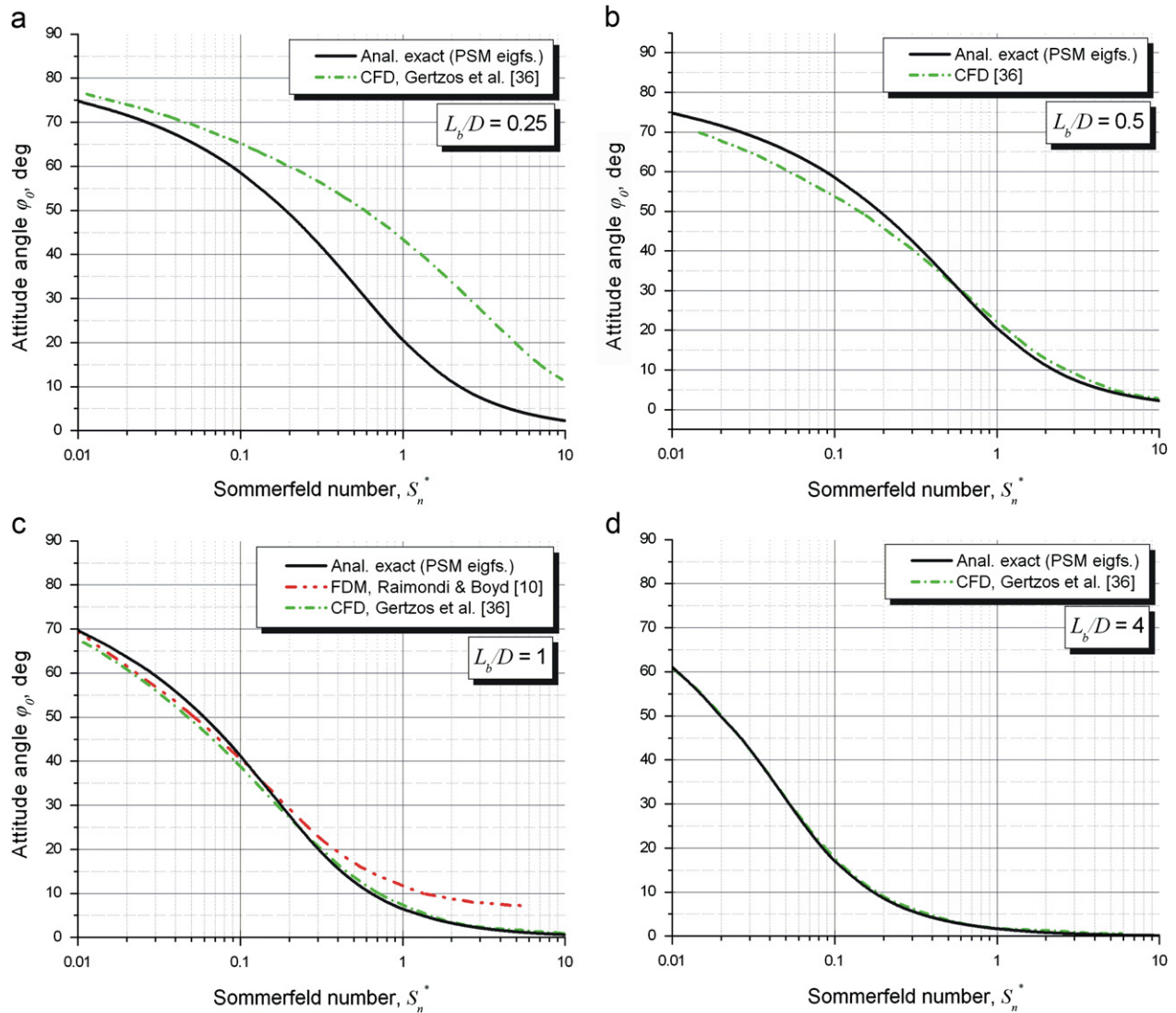


Fig. 7. Evaluation of the minimum fluid film thickness location as a function of Sommerfeld number for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$ using the analytical solutions and comparison with results from numerical solutions.

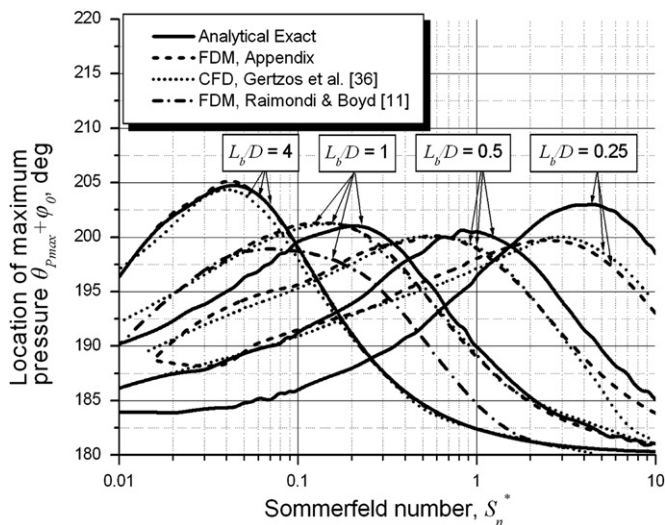


Fig. 8. Evaluation of the circumferential location of the maximum pressure ($\theta_{P_{max}} + \varphi_0$) at the axial centre of the bearing ($x=0$) as a function of the Sommerfeld number, for various L_b/D ratios using the exact analytical solution and numerical solution.

The approximate analytical solutions yield results in between those of FDM solution and those of the exact analytical solution. The discrepancy of the angle of maximum pressure notified between the analytical solutions is exclusively due to the Reynolds homogeneous solution that yields different “homogeneous pressure” profile near the axial centre of the bearing [21]. The convergence between the FDM and the CFD solution is very good and only at the case of L/D ratio of 1, where a FDM solution from the literature is also presented [11], a discrepancy of 3° is notified because in [11], and in the current result, the Reynolds BC is used instead of Gumbel BC, described as Type 2 BC [11]. As an additional remark in Fig. 8, a slight variation of the curve of the exact analytical solution is presented for the L/D ratios of 0.25 and 0.5 at around the Sommerfeld number values of 0.07 and 0.8 respectively. The reason for this slight variation is that evaluation of the angle of the maximum pressure is connected with the evaluation of the equilibrium position which is calculated using a 2D Newton–Raphson method. In some cases of Sommerfeld number only 3 or 4 iterations of the NRM are enough but in others, a number of 8 or more iterations are demanded. For avoiding time consumption, the iterations where set of constant value equal to 4, so in some cases this slight variation of less than 0.3° is notified. The curves of Fig. 8 conclude the discrepancies at

the two magnitudes, of this of the equilibrium position and this of the angle of maximum pressure with reference to the beginning of the fluid film pressure. Thus, two discrepancies are superimposed, this of equilibrium position coordinates and this of maximum pressure location. Both of them have to do with the use of the solution analytical, or numerical, therefore the discrepancies in the curves of Fig. 8 can be explained as an obvious result of the different theory of pressure calculation.

The location of the maximum pressure through the axial direction of the bearing is evaluated in order to obtain information about the pressure distribution. The locus of the maximum pressure is evaluated by solving the Eq. (4.2) using a Newton–Raphson method in the two variables of θ and x .

$$\left. \frac{\partial}{\partial \theta} \left(\varphi(\theta) + \sum_{i=1}^{\infty} (\sigma_i \bar{f}_i(\theta) m_i(x)) \right) \right|_{\theta = \theta_{pmax}} = 0 \quad (4.2)$$

The L/D ratios of 0.25, 0.5, 1 and 4 are incorporated in the evaluation and the three cases of loading that correspond to Sommerfeld number value of 0.01, 0.1 and 10 are investigated. The results using the three analytical and the FDM numerical solution (see Appendix A) are presented in Fig. 9. A first rough notification about the location of the maximum pressure comparing

the different L/D ratios and the different loads is that the maximum pressure locus gets nearer to the end of the π film as the load gets higher and also that the results regarding the different solutions converge more to each other as the L/D ratio gets higher. These notifications about the pressure distribution have been also made in the past literature but here they are mentioned in order to qualitatively validate the analytical solution and to reproduce significant measures analytically instead of numerically. The numerical result that is presented in Fig. 9 represents the numerical solution using FDM and is plotted in the charts of Fig. 9 as a reference to the numerical results. The comparison of the analytical results with the numerical result of FDM yields qualitative and quantitative differences. The differences become higher as the L/D ratio of the bearing becomes lower. For example, in the lightly loaded short bearing the exact analytical solution gives a maximum pressure locus with almost 10° discrepancy from the result of the numerical solution. In the same case of L/D ratio, and in the heavily loaded bearing, the corresponding difference becomes almost 2° . In all other cases of L/D ratio the differences become minimal or none. It is quite interesting to notify that in all cases of load and geometry the analytically evaluated maximum pressure locus is presented with a slight bending towards the end of the π film (the line of 180° in the charts of Fig. 9) as the axial coordinate of the locus

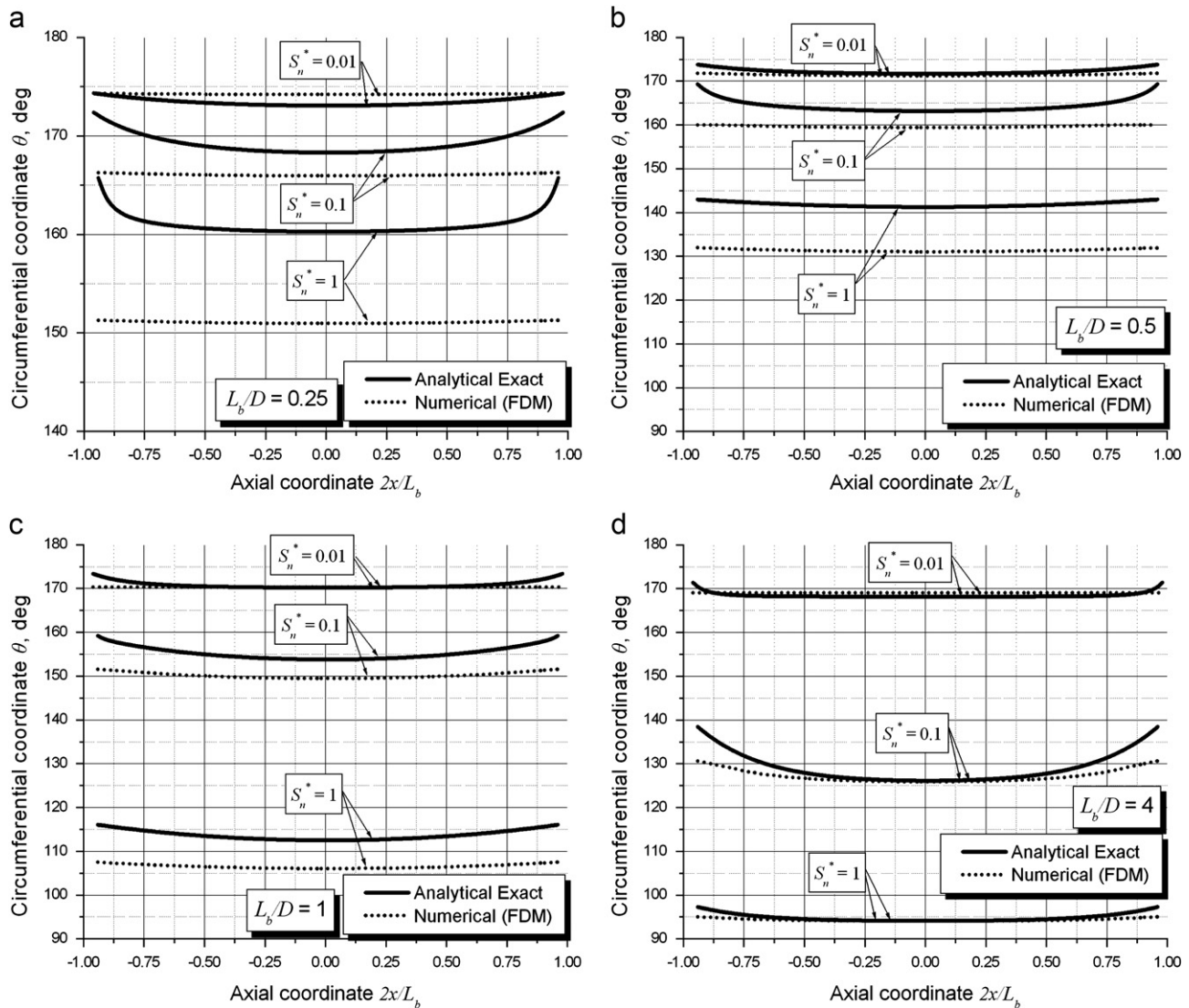


Fig. 9. Evaluation of the maximum pressure (θ_{pmax}) locus through the axial direction of the bearing for three different Sommerfeld number values, and for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$, using the analytical solutions and the FDM numerical solution.

becomes closer to the bearings ends (faces), with this graphical bending to be more intense in the case of the exact analytical solution. The locus of the maximum pressure that is evaluated analytically tends to approach the fluid film boundary of zero pressure at the angle of 180° (Gümbel B.C.) yielding more information about the quality of the pressure distribution.

5. Analytical evaluation of the stiffness and damping coefficients of the plain finite journal bearing

In this section the analytical expressions for the stiffness and damping coefficients of the plain finite journal bearing are evaluated. The partial derivatives of the journal bearing impedance forces given as a result of the exact analytical and the approximate analytical solutions are evaluated with respect to displacements in order to give the direct and the cross coupled stiffness coefficients and with respect to the journal centre velocities in order to give the direct and the cross coupled damping coefficients. The evaluation of the stiffness and damping coefficients is done in some chosen values of displacement y_0 and z_0 correspondent to the equilibrium root locus in the domain of $0.01 < S_n^* < 10$. The analytical expression of the fluid film pressure distribution as in Eq. (2.31) with the use also of the analytical

expressions for the impedance forces as in Eqs. (3.3) and (3.4) can yield the closed form expressions for the stiffness and the damping coefficients as shown in Eqs. (5.1) and (5.2) correspondingly. In Eq. (5.1) the values of the journal velocities $\dot{y}_0$ and $\dot{z}_0$ are set zero in order to correspond to static conditions.

The stiffness coefficients are calculated from the formulas in Eq. (5.1) for some cases of the bearing length to diameter ratio and they are presented in Figs. 10 and 11 in comparison with two other methods of bearing coefficients evaluation, the one presented from Lund [26] and the other being the numerical derivation after solving the Reynolds equation using FDM (see Appendix A).

$$K_{YY} = \frac{\partial F_Y}{\partial y} = \frac{y_0}{e} \frac{\partial F_r}{\partial y} + \frac{1}{e} F_r + \frac{z_0}{e} \frac{\partial F_t}{\partial y} \\ = \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial y} \cos(\theta) R dx d\theta + \frac{1}{e} F_r \\ + \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial y} \sin(\theta) R dx d\theta \quad (5.1a)$$

$$K_{YZ} = \frac{\partial F_Y}{\partial z} = \frac{y_0}{e} \frac{\partial F_r}{\partial z} + \frac{1}{e} F_r + \frac{z_0}{e} \frac{\partial F_t}{\partial z} \\ = \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial z} \cos(\theta) R dx d\theta + \frac{1}{e} F_t$$

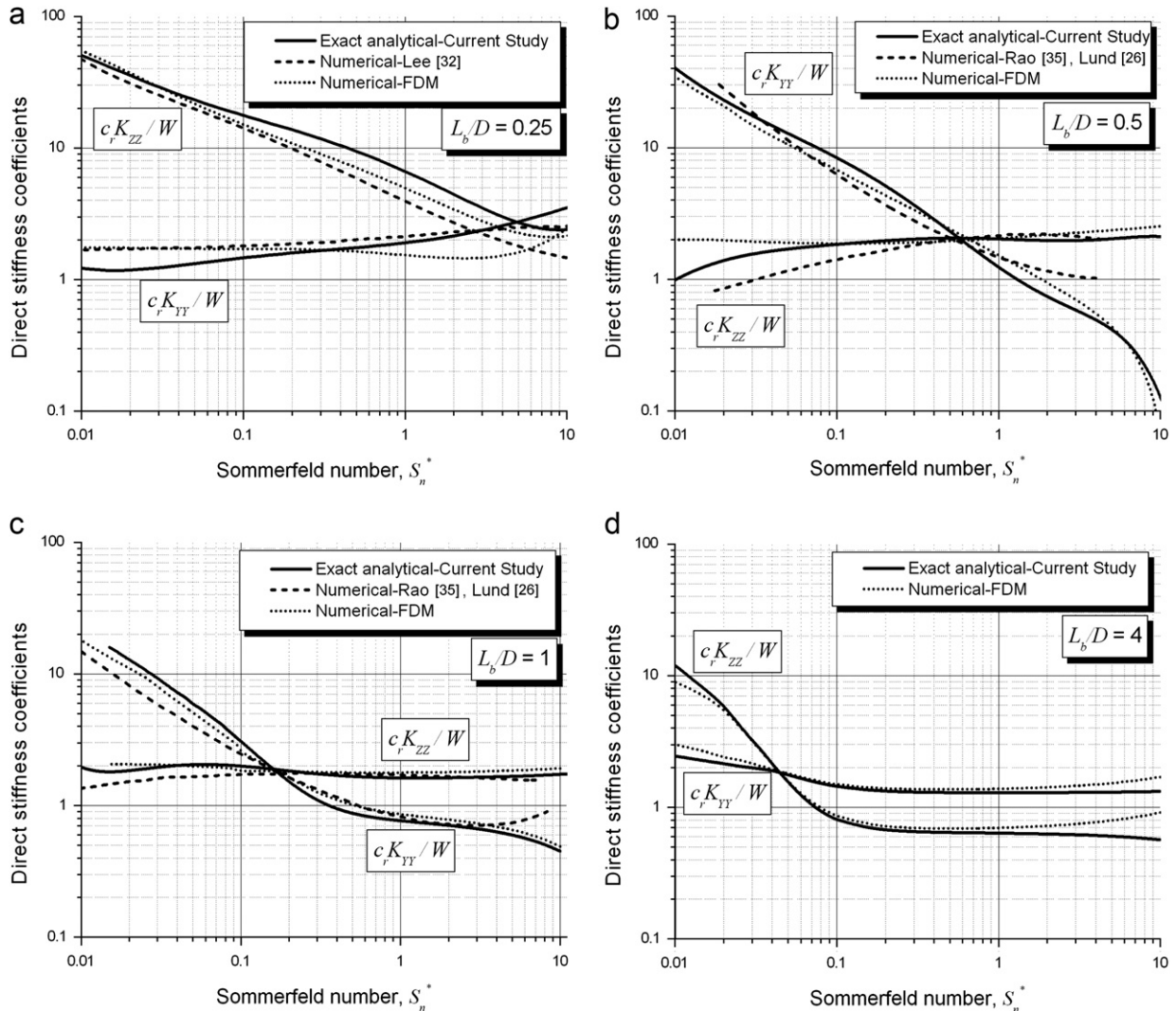


Fig. 10. Evaluation of the direct stiffness coefficients as a function of Sommerfeld number for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$ using the analytical solution and comparison with results from numerical solutions.

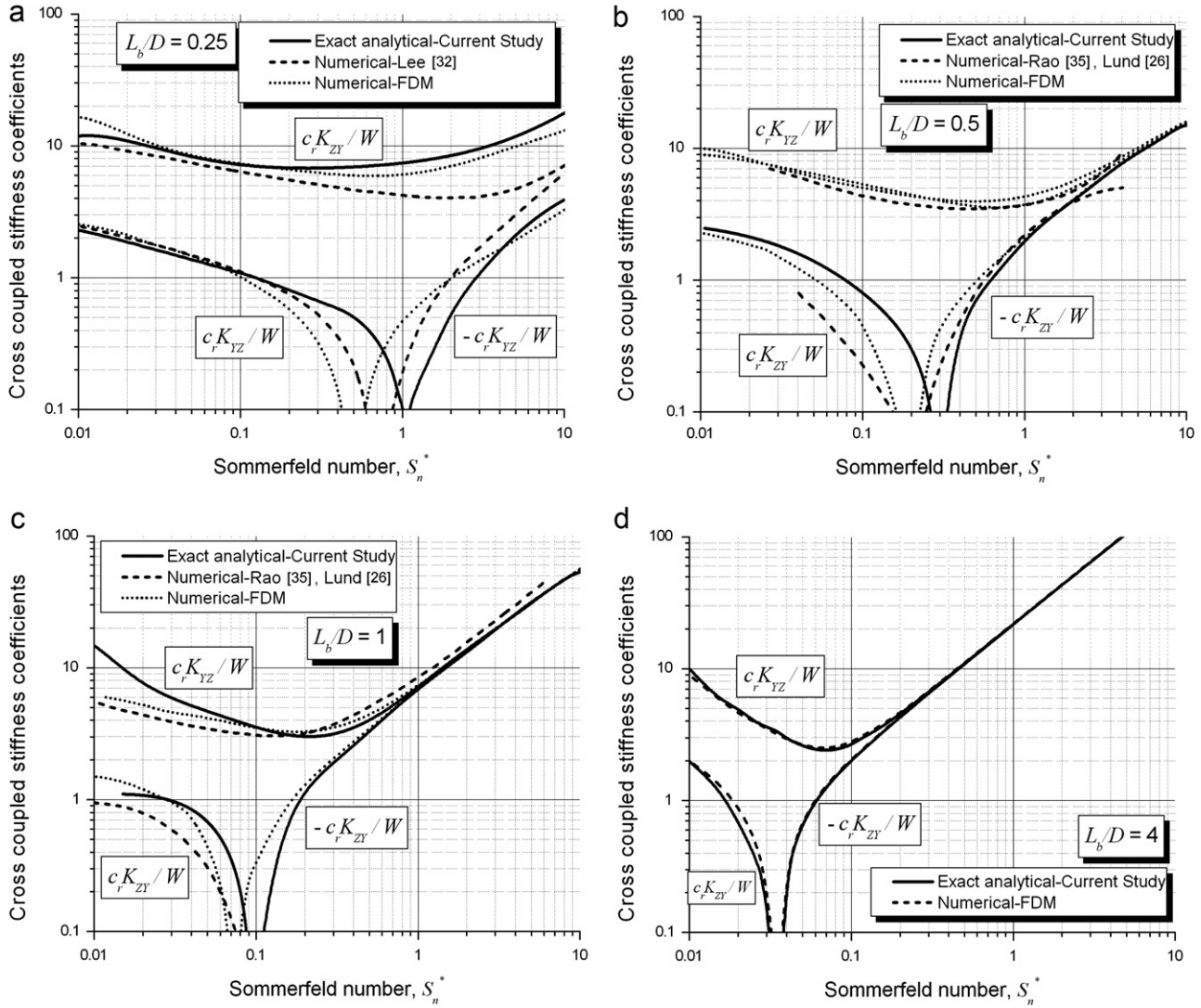


Fig. 11. Evaluation of the cross coupled stiffness coefficients as a function of Sommerfeld number for (a) $L_b/D=0.25$, (b) $L_b/D=0.5$, (c) $L_b/D=1$, and (d) $L_b/D=4$ using the analytical solution and comparison with results from numerical solutions.

$$+ \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial z} \sin(\theta) R dx d\theta \quad (5.1b)$$

$$K_{ZY} = \frac{\partial F_Z}{\partial y} = \frac{1}{e} F_t + \frac{y_0}{e} \frac{\partial F_t}{\partial y} - \frac{z_0}{e} \frac{\partial F_r}{\partial y} = \frac{1}{e} F_t + \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial y} \sin(\theta) R dx d\theta - \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial y} \cos(\theta) R dx d\theta \quad (5.1c)$$

$$K_{ZZ} = \frac{\partial F_Z}{\partial z} = \frac{y_0}{e} \frac{\partial F_t}{\partial z} - \frac{1}{e} F_r - \frac{z_0}{e} \frac{\partial F_r}{\partial z} = \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial z} \sin(\theta) R dx d\theta - \frac{1}{e} F_r - \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial z} \cos(\theta) R dx d\theta \quad (5.1d)$$

The partial derivatives of pressure that are incorporated in Eq. (5.1) are presented in Eq. (5.2).

$$\frac{\partial P(x, \theta)}{\partial y} = \frac{\partial \varphi}{\partial y} - \sum_{i=1}^{\infty} \left(\frac{\partial}{\partial y} \int_0^{2\pi} R^2 h^3 \bar{f}_i \frac{\partial \varphi}{\partial y} d\theta \right) \bar{f}_i(\theta) m_i(x) \quad (5.2a)$$

$$\frac{\partial P(x, \theta)}{\partial z} = \frac{\partial \varphi}{\partial z} - \sum_{i=1}^{\infty} \left(\frac{\partial}{\partial z} \int_0^{2\pi} R^2 h^3 \bar{f}_i \frac{\partial \varphi}{\partial z} d\theta \right) \bar{f}_i(\theta) m_i(x) \quad (5.2b)$$

The damping coefficients are evaluated using the formulas of Eq. (5.3) and they are presented in Figs. 12 and 13 in comparison with numerical results.

$$C_{YY} = \frac{\partial F_Y}{\partial \dot{y}} = \frac{y_0}{e} \frac{\partial F_r}{\partial \dot{y}} + \frac{z_0}{e} \frac{\partial F_t}{\partial \dot{y}} = \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{y}} \cos(\theta) R dx d\theta + \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{y}} \sin(\theta) R dx d\theta \quad (5.3a)$$

$$C_{YZ} = \frac{\partial F_Y}{\partial \dot{z}} = \frac{y_0}{e} \frac{\partial F_r}{\partial \dot{z}} + \frac{z_0}{e} \frac{\partial F_t}{\partial \dot{z}} = \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{z}} \cos(\theta) R dx d\theta + \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{z}} \sin(\theta) R dx d\theta \quad (5.3b)$$

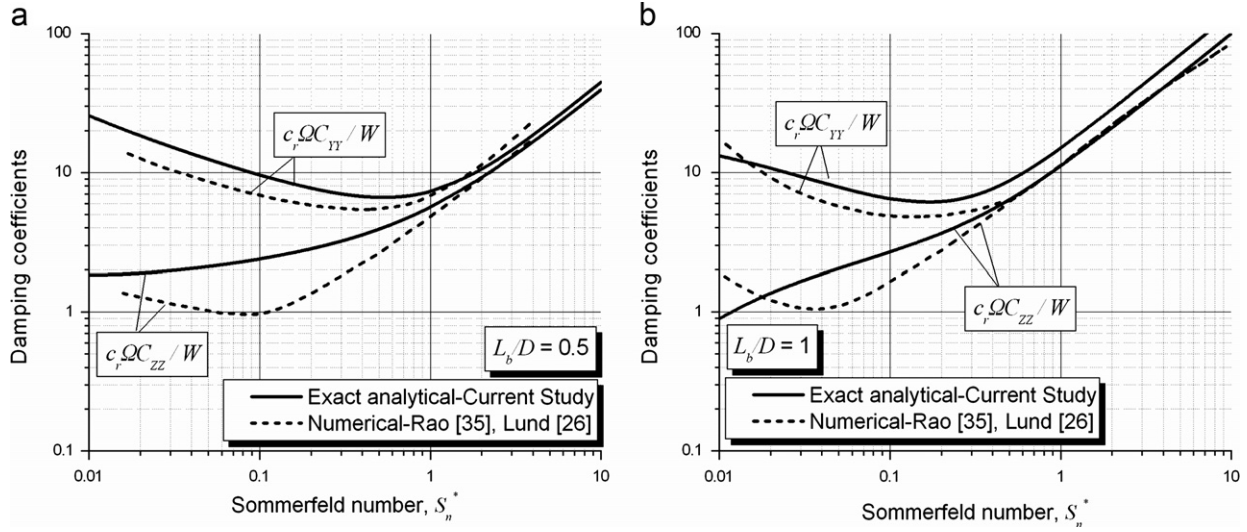


Fig. 12. Evaluation of the direct damping coefficients as a function of Sommerfeld number for (a) $L_b/D=0.5$, and (b) $L_b/D=1$, using the analytical solution and comparison with results from the numerical solution.

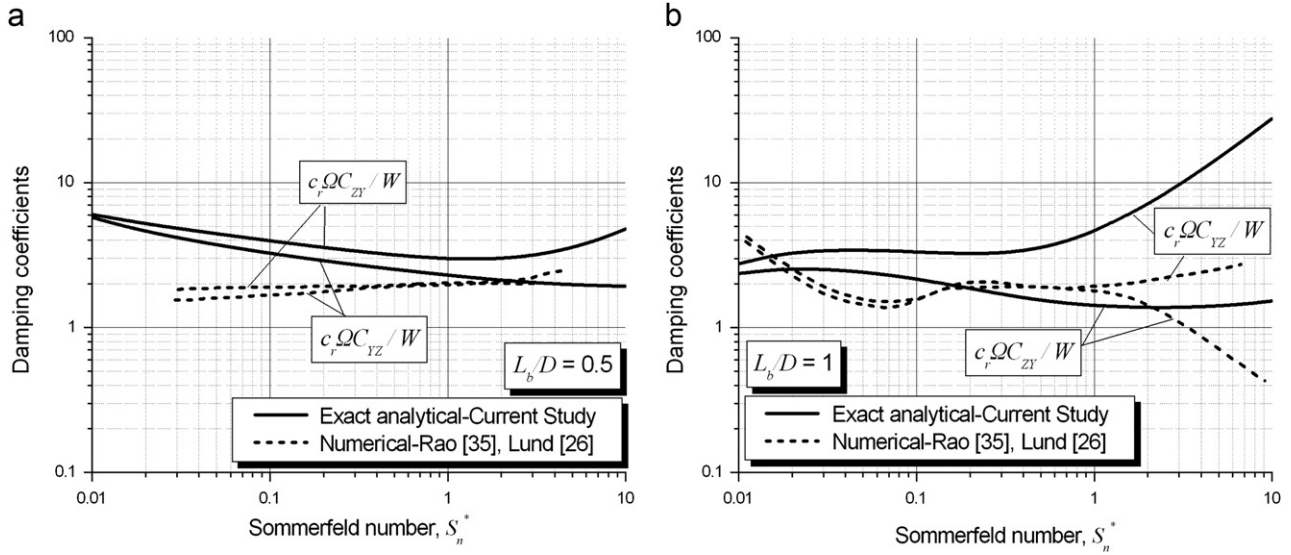


Fig. 13. Evaluation of the cross coupled damping coefficients as a function of Sommerfeld number for (a) $L_b/D=0.5$, and (b) $L_b/D=1$, using the analytical solution and comparison with results from the numerical solution.

$$\begin{aligned}
 C_{ZY} &= \frac{\partial F_Z}{\partial \dot{y}} = \frac{y_0}{e} \frac{\partial F_t}{\partial \dot{y}} - \frac{z_0}{e} \frac{\partial F_r}{\partial \dot{y}} \\
 &= \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{y}} \sin(\theta) R dx d\theta \\
 &\quad - \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{y}} \cos(\theta) R dx d\theta
 \end{aligned} \quad (5.3c)$$

$$\begin{aligned}
 C_{ZZ} &= \frac{\partial F_Z}{\partial \dot{z}} = \frac{y_0}{e} \frac{\partial F_t}{\partial \dot{z}} - \frac{z_0}{e} \frac{\partial F_r}{\partial \dot{z}} \\
 &= \frac{y_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{z}} \sin(\theta) R dx d\theta \\
 &\quad - \frac{z_0}{e} \int_0^\pi \int_{-L_b/2}^{L_b/2} \frac{\partial P(x, \theta)}{\partial \dot{z}} \cos(\theta) R dx d\theta
 \end{aligned} \quad (5.3d)$$

The partial derivatives of pressure incorporated in Eq. (5.3) are presented in Eq. (5.4).

$$\frac{\partial P(x, \theta)}{\partial \dot{y}} = \frac{\partial \varphi}{\partial \dot{y}} - \sum_{i=1}^{\infty} \left(\frac{\partial}{\partial \dot{y}} \int_0^{2\pi} R^2 h^3 \bar{f}_i \frac{\partial \varphi}{\partial \dot{y}} d\theta \right) \bar{f}_i(\theta) m_i(x) \quad (5.4a)$$

$$\frac{\partial P(x, \theta)}{\partial \dot{z}} = \frac{\partial \varphi}{\partial \dot{z}} - \sum_{i=1}^{\infty} \left(\frac{\partial}{\partial \dot{z}} \int_0^{2\pi} R^2 h^3 \bar{f}_i \frac{\partial \varphi}{\partial \dot{z}} d\theta \right) \bar{f}_i(\theta) m_i(x) \quad (5.4b)$$

In Figs. 10–13 is clear that there are discrepancies between the analytical and the numerical results in both stiffness and damping coefficients. The clearest reason for the discrepancies notified has to do with the differences that were presented in the magnitude of pressure (see Fig. 2). The pressure is the outcome of the solution (numerical or analytical) and those differences notified in Fig. 2 are transferred to the forces through integration and finally to the coefficients through derivation. In the magnitudes of the stiffness coefficients one can see that the analytical and numerical results are in a very good agreement as show clearly in Figs. 10 and 11. However, in the magnitudes of damping coefficients, the agreement is not of the same level as in stiffness coefficients. The reason for this is the influence of the eccentricity rate of change $\dot{e}$ in the pressure distribution. In the analytical solution the eccentricity rate of change influences the pressure through the particular solution only, see Eq. 2.13. As it was shown

in [21] the influence of the eccentricity rate of change $\dot{e}$ in the pressure distribution is of high sensitivity, this is actually well known since the property of journal bearings to develop high pressures because of the developed velocity of the journal makes them very applicable in systems that high damping is needed. This sensitivity of the particular solution and therefore of the resulting pressure to the eccentricity rate of change $\dot{e}$ is the reason for the discrepancies presented in Figs. 12 and 13. In the case of the numerical solutions, the influence of $\dot{e}$ is also of high sensitivity although in [21] is presented that the pressure evaluated with the analytical solution is slightly higher when the $\dot{e}$ is incorporated. The higher the $\dot{e}$ gets, the higher the discrepancy in the pressure is. Additionally, the coefficients obtained by direct analytical differentiation of the pressure (see Eq. (5.4)) with respect to $\dot{y}$ or $\dot{z}$ are obtained through a completely different path from this of the direct numerical differentiation or the pressure or of the perturbation solution of the Reynolds [41]. Actually, according to [41], the coefficients may be computed by direct numerical differentiation, but as this is inherently inaccurate they should preferably be obtained from a perturbation solution of Reynolds equation.

6. Conclusions, contribution and future concepts

The current work uses an exact analytical solution of the Reynolds equation for the finite journal bearing lubrication in order to produce results for the important design parameters and operational characteristics of the full cylindrical plain finite journal bearing. The analytical function of the pressure distribution is treated properly in order to extract analytical functions about the finite journal bearing impedance forces which are expressed in closed form expressions for three cases of analytical solution: (a) the exact analytical using Power Series Method, (b) the approximate analytical using Bessel eigenfunctions, and (c) the approximate analytical using sinusoidal eigenfunctions. The three methods of solution yield similar impedance forces but the approximate analytical solutions yield much more compact forces formulas than the exact analytical, especially in the case that the used boundary condition is this of Gumbel, as it was used in this paper.

The evaluation of the bearing parameters and characteristics made under the three aforementioned cases of solution yields very good agreement in the parameter of load carrying capacity. Actually, the three different solutions yield eccentricity ratios with very good agreement to each other and also with very good agreement with those yielded from numerical methods in the literature. This observation goes for the four different L/D ratios of 0.25, 0.5, 1 and 4 that are treated in the current work.

The position of the minimum fluid film thickness shows also very good agreement between the analytical results and with the numerical results especially when the L/D ratio becomes 1 and 4. For the L/D ratio of 0.5 some small discrepancies is notified between the analytical solutions and the analytical and the numerical. This discrepancy becomes up to 20° for the case of L/D ratio of 0.25.

The angular location of the maximum pressure through the axial direction of the finite bearing is evaluated analytically and numerically (using FDM) for three different cases of loading conditions (high, medium and low Sommerfeld number) and for the four aforementioned cases of L/D ratio. The very good agreement is notified in the case of L/D ratio of 4 for the three cases of loading. In the case of L/D ratio of 1 the discrepancy between the exact analytical result and the numerical result reaches up to 5° in the case of light loading while no discrepancy is notified in the case of high loading except some slight

“bending” of the analytically evaluated locus near the bearing axial ends. In the case of L/D of 0.5 the maximum discrepancy reaches the 10° when the load is light and there is almost absolute convergence when the load is high. In the short bearing of L/D ratio of 0.5 the maximum discrepancy between exact analytical and FDM numerical results reaches also the 10° for the light load and almost the 2° when the load is high.

The bearing characteristics of the eight stiffness and damping coefficients are evaluated analytically with direct differentiation of the impedance forces with respect to journal centre displacements and velocities. The benefit of the analytical solution is that the pressure function can be directly differentiated and integrated as well with the outcome to be always expressible as closed form expressions. Considering the up to nowadays methods for evaluating finite bearing characteristics, the current possibility of the analytical evaluation of them contributes in this concept a compact and simple way of obtaining those important parameters for the design of a rotor bearing system by using closed form formulas. The stiffness coefficients are compared to those obtained numerically in the classic literature and also to those obtained analytically in the literature under short and long bearing approximation. The analytical formulas presented in this work for the eight stiffness and damping coefficients for the finite journal bearing follow very good agreement in the most range of Sommerfeld number that is considered to vary from 0.01 (heavy load) up to 10 (light load) while the differences can be explained from the fact that the numerical procedures follow different methods of extracting the bearing coefficients than this of the simple direct differentiation of the analytical functions of the forces.

The following concepts can be highlighted as the contribution of the current work in the concept of finite journal bearing design:

- (a) The presentation of an exact analytical solution of the Reynolds equation for the lubrication of the plain finite journal bearing and the expression of the pressure distribution with an analytical function that was the base for the rest output of the bearing design parameters.
- (b) The definition of the impedance forces of the finite journal bearing under the exact analytical solution and also under two approximate analytical solutions with the assumption of linearly varying fluid film thickness function that was suggested in the past.
- (c) The definition of the stiffness and damping coefficients of the finite journal bearing with the expression of analytical functions that incorporate operational, physical and geometrical parameters of the bearing. These functions are presented for the three cases of the analytical treatment.

The idea of giving analytical solutions in equations of high applicability and usage, such as this of Reynolds, is a matter that has also to do with the philosophy under which the problem of journal bearing lubrication is treated. This means that the fact of solving analytically the lubrication problem of the finite journal bearing perhaps can yield further knowledge in understanding the concept of the journal bearing in its sense of nature. The numerical processes are those that are mostly used but it would be very useful in the future to be able to express the pressure function and the forces formulas for the plain finite journal bearing much more simply, without even the use of Power Series, contributing in the direction of bearing design as much as the analytical formulas of the Short/Long bearing contributed up to know in the engineering community. Many other parameters of bearing design are left for analytical evaluation in future works and together with possible further mathematical treatment, the expressions of the bearing design parameters can reach a more

compact and applicable form even for different cases of boundary conditions for the pressure distribution.

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Appendix A

The numerical results presented through this paper under the reference of FDM method are produced with the following algorithm for evaluation of the fluid film pressure using the extensively used finite difference method for solving a PDE. The algorithm is presented in this appendix in short in order to give more information about the numerical results that often appeared in the charts.

The Reynolds equation given in Eq. (2.1) is transformed as in Eq. (A.1) in order to correspond to a finite difference grid of the circumferential and axial direction of the bearing. The dimension of the grid is 45×15 with respect to the circumferential and axial direction correspondingly. The boundary condition for the pressure profile corresponds to this of Gumbel (half Sommerfeld) and it is expressed using the Eq. (A.2).

$$\frac{h_j^3}{6\mu} \frac{P_{i+1,j} - 2P_{i,j} + P_{i-1,j}}{\delta x^2} + \frac{1}{R^2} \left(\frac{-e h_j^2 \sin(\theta_j) P_{i,j+1} - P_{i,j-1}}{2\mu} + \frac{h_j^3}{6\mu} \frac{P_{i,j+1} - 2P_{i,j} + P_{i,j-1}}{\delta \theta^2} \right) =, \quad i = 1, 2, \dots, 14, \quad j = 1, 2, \dots, 44$$

$$= -(\Omega - 2\dot{\phi})e \sin(\theta_j) + 2\dot{e} \cos(\theta_j) \quad (\text{A.1})$$

$$P_{0,j} = 0 \text{ and } P_{15,j} = 0, \quad j = 0, 1, 2, \dots, 45$$

$$P_{i,0} = 0 \text{ and } P_{i,45} = 0, \quad i = 0, 1, 2, \dots, 15 \quad (\text{A.2})$$

In Eq. (A.1) the fluid film thickness is expressed as $h_j = c_r + e \cos(\theta_j)$, the circumferential coordinate as $\theta_j = j\delta\theta$ and the intervals $\delta\theta$ and δx as $\delta\theta = \pi/45$ and $\delta x = L_b/15$. Also the axial coordinate is defined as $x_i = -L_b/2 + i\delta x$. The 44×14 linear algebraic system is solved using a Gauss elimination algorithm.

The resulting data of the pressure values in every point of the grid is interpolated with a 7th order BSpline interpolation and thus the function of pressure distribution is defined analytically in every point of the geometrical domain. The analytical expansion using BSpline surface gives the possibility of evaluating the partial derivative of the pressure with respect to θ and obtain the locations of the maximum pressure as shown in Figs. 8 and 9.

For the numerical evaluation of the stiffness and damping coefficients the resulting fluid film forces are evaluated as in Eq. (A.3) by setting $\dot{e} = 0$ and $\dot{\phi} = 0$ and thus the locations of equilibrium are evaluated as y_0 and z_0 (see Fig. 1).

$$F_r = \sum_{i=1}^{14} \sum_{j=1}^{44} (P_{i,j} R \cos(\theta_j) \delta\theta \delta x) \quad \text{and} \quad F_Y = F_r \frac{y}{e} + F_t \frac{z}{e}$$

$$F_t = \sum_{i=1}^{14} \sum_{j=1}^{44} (P_{i,j} R \sin(\theta_j) \delta\theta \delta x) \quad \text{and} \quad F_Z = F_t \frac{y}{e} - F_r \frac{z}{e} \quad (\text{A.3})$$

A first perturbation is applied in the vertical and horizontal displacements as $y = y_0 + \delta y_1$ and $z = z_0 + \delta z_1$ and the forces are evaluated as F_{Y_1} and F_{Z_1} . A second perturbation of $y = y_0 + \delta y_2$ and $z = z_0 + \delta z_2$ is applied and the forces in the new equilibrium

position become F_{Y_2} and F_{Z_2} . The differences in the forces due to the perturbations are expressed as $\delta F_Y = F_Y - W_Y$ and $\delta F_Z = F_Z - W_Z$. By expressing the bearing impedance forces as in Eq. (A.4) [35] the stiffness coefficients are evaluated as in Eq. (A.5)

$$\delta F_Y = -K_{YY} \delta y - K_{YZ} \delta z, \quad \delta F_Z = -K_{ZZ} \delta z - K_{ZY} \delta y \quad (\text{A.4})$$

$$K_{YY} = -\frac{\delta F_{Y_2} \delta z_1 + \delta F_{Y_1} \delta z_2}{-\delta y_2 \delta z_1 + \delta y_1 \delta z_2}, \quad K_{ZZ} = -\frac{\delta F_{Z_2} \delta y_1 + \delta F_{Z_1} \delta y_2}{-\delta y_2 \delta z_1 + \delta y_1 \delta z_2},$$

$$K_{YZ} = -\frac{\delta F_{Y_2} \delta y_1 + \delta F_{Y_1} \delta y_2}{-\delta y_2 \delta z_1 + \delta y_1 \delta z_2}, \quad K_{ZY} = -\frac{\delta F_{Z_2} \delta z_1 + \delta F_{Z_1} \delta z_2}{-\delta y_2 \delta z_1 + \delta y_1 \delta z_2} \quad (\text{A.5})$$

In order to evaluate the damping coefficients the perturbations are applied in the variables of $\dot{y}$ and $\dot{z}$ as $\dot{y} = \delta \dot{y}_1$ and $\dot{z} = \delta \dot{z}_2$ in order to evaluate the first set of forces F_{Y_1} and F_{Z_1} , and as $\dot{y} = \delta \dot{y}_2$ and $\dot{z} = \delta \dot{z}_2$ in order to evaluate the second set of forces F_{Y_2} and F_{Z_2} . The differences in the forces are now defined as in Eq. (A.6) [35] and the damping coefficients are evaluated as in Eq. (A.7)

$$\delta F_Y = -C_{YY} \delta \dot{y} - C_{YZ} \delta \dot{z}, \quad \delta F_Z = -C_{ZZ} \delta \dot{z} - C_{ZY} \delta \dot{y} \quad (\text{A.6})$$

$$C_{YY} = -\frac{\delta F_{Y_2} \delta \dot{z}_1 + \delta F_{Y_1} \delta \dot{z}_2}{-\delta \dot{y}_2 \delta \dot{z}_1 + \delta \dot{y}_1 \delta \dot{z}_2}, \quad C_{ZZ} = -\frac{\delta F_{Z_2} \delta \dot{y}_1 + \delta F_{Z_1} \delta \dot{y}_2}{-\delta \dot{y}_2 \delta \dot{z}_1 + \delta \dot{y}_1 \delta \dot{z}_2},$$

$$C_{YZ} = -\frac{\delta F_{Y_2} \delta \dot{y}_1 + \delta F_{Y_1} \delta \dot{y}_2}{-\delta \dot{y}_2 \delta \dot{z}_1 + \delta \dot{y}_1 \delta \dot{z}_2}, \quad C_{ZY} = -\frac{\delta F_{Z_2} \delta \dot{z}_1 + \delta F_{Z_1} \delta \dot{z}_2}{-\delta \dot{y}_2 \delta \dot{z}_1 + \delta \dot{y}_1 \delta \dot{z}_2} \quad (\text{A.7})$$

The perturbations in the displacements are chosen as $\delta y_1 = 10^{-3} c_r$, $\delta y_2 = 2 \times 10^{-3} c_r$, $\delta z_1 = 10^{-3} c_r$, $\delta z_2 = 1.5 \times 10^{-3} c_r$ and in the velocities as $\delta \dot{y}_1 = 10^{-6} \Omega R$, $\delta \dot{y}_2 = 2 \times 10^{-6} \Omega R$, $\delta \dot{z}_1 = 10^{-6} \Omega R$, $\delta \dot{z}_2 = 1.5 \times 10^{-6} \Omega R$. Different values of perturbation were also tried in order to identify convergence. The values of the Sommerfeld number in the numerical (and in the analytical) evaluation of the stiffness and damping coefficients shown in Figs. 10–13 where 55 and specifically the values 10, 9.5, 9, 8.5...1, 0.95, 0.9, 0.85...0.1, 0.095, 0.09, 0.085...0.01. An Spline interpolation of the aforementioned values of data was used in order to produce the diagrams shown in Figs. 10–13.

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